# Estimating Infectious Disease Importation Risk during the 2026 FIFA World Cup

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**Abstract**

**Background.** The 2026 FIFA World Cup will bring an estimated 1–5 million international visitors to 11 US host cities between June 11 and July 19, 2026—the largest tournament in history. Large-scale international gatherings accelerate importation of infectious diseases from diverse source populations. Advance estimation of importation risk is essential for public health preparedness and surveillance prioritization.

**Methods.** We developed a Poisson importation framework applied to five diseases (dengue fever, influenza, malaria, measles, and pertussis) across the 11 US venue cities. Three nested travel models of increasing resolution were constructed: a baseline model using routine June 2024 arrival data; a World Cup–adjusted model incorporating projected visitor growth factors; and a schedule-driven model routing WC fans to specific cities based on match assignments. WHO incidence and BTS T-100 routing fractions were combined with Monte Carlo uncertainty propagation (5,000 Uniform draws on under-reporting and travel-while-infectious parameters) to yield median importation estimates with 95% uncertainty intervals.

**Results.** Dengue posed the highest importation risk at most venue cities under the schedule-driven model (median $\Lambda > 10$ expected importations from Brazil alone; 95%

uncertainty interval 5.9–33.1), robust across the full literature-supported parameter range; Atlanta was the exception, where malaria probability exceeded dengue, driven by direct travel from West and Central African nations. Influenza ranked second at most cities, coinciding with the Southern Hemisphere winter peak. Pertussis showed broad geographic spread but carries the widest relative uncertainty, as the assumed detection rate sits at the upper bound of the literature range. Background tourism accounted for the dominant share of total importation risk; the World Cup fan increment contributed approximately 8.3% of projected arrivals for WC-qualified nations.

**Conclusions.** This Poisson importation framework, built entirely from publicly available data, provides reproducible importation risk estimates for mass gathering events. The framework extends to additional diseases, cities, and gatherings, offering a transparent baseline complementary to proprietary modeling systems.

# Introduction

Mass gathering events concentrate large numbers of international visitors in a short period, creating conditions for accelerated importation of infectious diseases from diverse source populations. The 2026 FIFA World Cup—the largest in the tournament's history, with 48 qualified nations—will draw visitors from every world region to 11 US host cities during June and July 2026, a period that coincides with peak dengue transmission in the Southern Hemisphere and with active global circulation of several vaccine-preventable diseases.

Estimating importation risk in advance is essential for surveillance prioritization and public health preparedness. Existing frameworks often rely on aggregate passenger volumes that obscure spatial and source-country heterogeneity, focus on a single disease, or do not account for the event-specific travel surge. Here we present a general, data-driven Poisson importation framework that addresses these gaps, applied to the 2026 FIFA World Cup as a case study.

The framework integrates multiple publicly available datasets—US Customs and Border Protection monthly arrivals (COR I-94), BTS T-100 airport routing data, NTTO travel projections, and WHO disease surveillance—into three nested models of increasing resolution. The three-model structure allows clean attribution: the step from Model 1 to Model 2 isolates the effect of the World Cup travel surge; the step from Model 2 to Model 3 reveals the added value of schedule-based fan routing. The approach is transparent, reproducible using only public data, and designed to be extensible to additional diseases and geographies.

# Methods

## Importation framework

We model importation risk as a Poisson process. The expected number of infectious travelers arriving at US venue city $v$ from country $c$ for disease $d$ is:

$$\lambda_{c,v,d} = N_{c,v} \cdot I_{c,d} \cdot p_d \tag{1}$$

where $N_{c,v}$ is the projected number of travelers from country $c$ to city $v$, $I_{c,d} = \rho_d \cdot C_{c,d}/P_c$ is the country-level disease incidence adjusted for under-reporting (with $C_{c,d}$ reported cases, $P_c$ population, and $\rho_d$ the under-reporting factor), and $p_d$ is the probability of traveling while infectious. Total importation intensity at each venue city is $\Lambda_{v,d} = \sum_c \lambda_{c,v,d}$, and the probability of at least one importation is $P(\geq 1) = 1 - e^{-\Lambda_{v,d}}$.

## Three-model hierarchy

We applied the framework under three travel models of increasing resolution (Table 1). All three models use the same BTS T-100 routing fractions $f_{c,v}$ (mean June 2023–2025) to distribute country-level arrivals across the 11 US venue cities, and the same country-level disease incidence estimates.

| **Model** | **Travel volume** | **Key feature** |
|---|---|---|
| Model 1: Baseline | COR June 2024 arrivals, no WC adjustment | Lower bound; isolates baseline risk from routine June tourism |
| Model 2: WC-adjusted | June 2026 projections via NTTO $\phi_c$ factors | Captures WC surge; $\phi_{\text{WC}} = 1.174$ for qualified nations, $\phi_{\text{base}} = 1.077$ for others |
| Model 3: Schedule-driven | Travel decomposed into WC-fan stream (match schedule) + background stream (T-100) | Spatial routing of fans to the specific cities where their team plays |

Table 1: Three-model hierarchy. All models use the same T-100 routing fractions and disease parameters; they differ only in how travel volumes are specified.

## Data sources and parameters

Travel volumes were derived from COR I-94 monthly arrivals by country of residence [1], scaled to June 2026 using NTTO international visitor forecasts [2] for the top-12 source markets, and to BTS T-100 International Segment data for city-level routing fractions [3]. Match schedules were obtained from official FIFA records [4]. Disease incidence was derived from WHO surveillance data: dengue from the WHO/PAHO global dengue dashboard [5], malaria from the World Malaria Report [6], measles and pertussis from the WHO Immunization Data portal [7], and influenza from the WHO FluNet / GISRS global sentinel surveillance system (ISO weeks 22–26, 2023–2025). Under-reporting factors and travel-while-infectious probabilities were taken from the literature [6, 8–11]. Key parameter values are: dengue ($\rho = 0.10$, $p_d = 0.50$), malaria ($\rho = 0.20$, $p_d = 0.30$), measles ($\rho = 0.60$, $p_d = 0.05$), pertussis ($\rho = 0.10$, $p_d = 0.70$), and influenza ($\rho = 0.10$, $p_d = 0.50$). Full details are given in Appendix A.

## Monte Carlo uncertainty propagation

Both $\rho_d$ (under-reporting factor) and $p_d$ (travel-while-infectious probability) are uncertain across one or more orders of magnitude in the literature (Table 3 in Appendix A). To propagate this uncertainty through the importation estimates, we drew 5,000 independent samples of each parameter from independent Uniform distributions calibrated to the literature ranges: $\rho_d \sim \text{Uniform}(\rho_{\min}, \rho_{\max})$ and $p_d \sim \text{Uniform}(p_{\min}, p_{\max})$. Uniform distributions were chosen for transparency; they make no assumption about the shape of uncertainty within the documented range.

Because $\Lambda_{v,d} = p_d \cdot \rho_d \cdot \sum_c (C_{c,d}/P_c) \cdot N_{c,v}$ is linear in the product $(p_d \cdot \rho_d)$, each Monte Carlo draw rescales the central estimate by a factor $(p_d^{(i)} \cdot \rho_d^{(i)})/(p_c \cdot \rho_c)$. We report the median and 95% interval (2.5th–97.5th percentile) across the 5,000 draws. Figures show bars at the median with whiskers spanning the 95% interval; the probability of at least one importation $P(\geq 1)$ is derived from the sampled $\Lambda$ distributions. Literature ranges used are: dengue ($\rho$: 0.06–0.26; $p$: 0.30–0.70), malaria ($\rho$: 0.10–0.35; $p$: 0.10–0.50), measles ($\rho$: 0.40–0.80; $p$: 0.02–0.10), pertussis ($\rho$: 0.01–0.10; $p$: 0.50–0.90), and influenza ($\rho$: 0.033–0.20; $p$: 0.30–0.70).

# Results

## Importation risk overview

Figure 1 shows the probability of at least one importation $[P(\geq 1)]$ for all five diseases across the 11 US venue cities under the schedule-driven model (Model 3). Dengue fever leads importation risk at most venue cities, with $P(\geq 1) > 0.99$ at Miami, New York, and Houston and ranging from 0.09 to 0.97 at the remaining cities. Atlanta is a notable exception: malaria reaches $P(\geq 1) > 0.99$—exceeding dengue (0.97)—driven by direct travel from West and Central African nations. Influenza ranks second at most cities, its risk amplified by coincidence with the Southern Hemisphere winter peak. Pertussis shows broad but moderate probability; outside Atlanta and New York, malaria risk is low and geographically concentrated. Measles consistently shows the lowest probability, driven by its severe prodrome suppressing travel during the infectious window ($p_d = 0.05$). These five diseases occupy distinct niches—by magnitude, geography, and uncertainty structure—as the following subsections detail. The equivalent heatmaps for the Baseline (Model 1) and WC-adjusted (Model 2) tiers are shown in Appendix B (Figures 6–7).

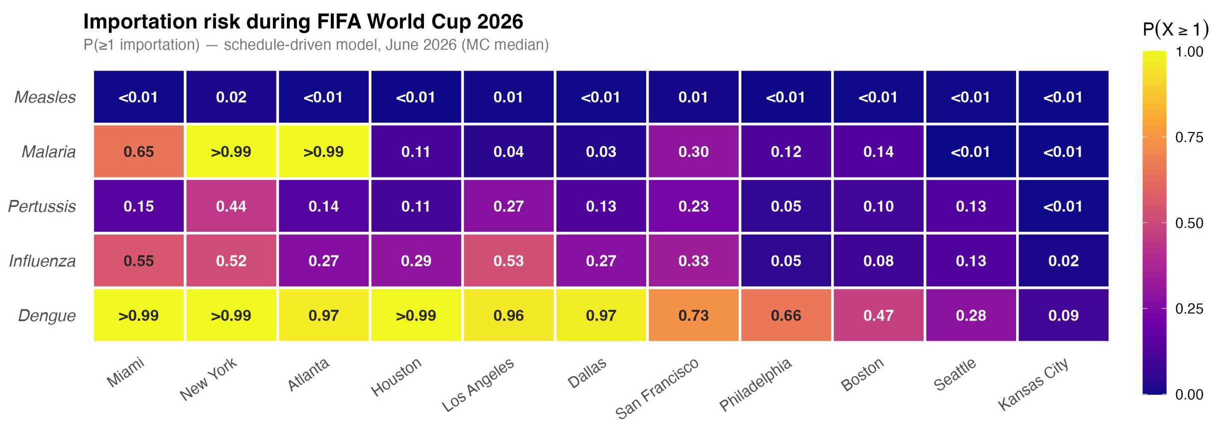


Figure 1: Importation risk overview for the FIFA World Cup 2026. Each cell shows $P(\geq 1$ importation) under the schedule-driven model (Model 3), June 2026, with Monte Carlo median from 5,000 parameter draws. Cities are ordered left to right by decreasing aggregate importation intensity; diseases are ordered top to bottom by decreasing overall probability. Dengue leads at most cities; malaria exceeds dengue at Atlanta ($P(\geq 1) > 0.99$ vs. 0.97). Measles risk is the lowest of the five diseases considered.

## Importation intensity and parameter uncertainty

Figure 2 shows the expected importation intensity ($\Lambda$, median and 95% Monte Carlo uncertainty interval) per city for each disease. For dengue, New York leads (median $\Lambda = 12.4$, 95% CI 6.1–30.8), followed by Los Angeles ($\Lambda = 11.8$, CI 5.8–29.2) and Miami ($\Lambda = 10.9$, CI 5.3–27.1), driven in all cases by Brazil's combination of high endemic burden and high June travel volume. The dengue CI extends substantially above the median because the central detection rate $\rho = 0.10$ lies near the lower end of the literature range (0.06–0.26): true importation intensity may be considerably higher than the central estimate.

Pertussis shows the opposite asymmetry. Its central $\rho = 0.10$ is the *upper* bound of its literature range (0.01–0.10), so the 95% CI extends far below the median. Pertussis estimates are best read as conservative upper bounds: at the lower end of the plausible detection range, expected importations could be as low as 7% of the median. Malaria carries a moderate, roughly symmetric CI. Measles and influenza uncertainties are modest relative to their medians. The directional asymmetry of each CI is visualised in Figure 5 in Appendix B: diseases with upward-skewed CIs (dengue, influenza) may be underestimated; diseases with downward-skewed CIs (pertussis) are conservative upper bounds.

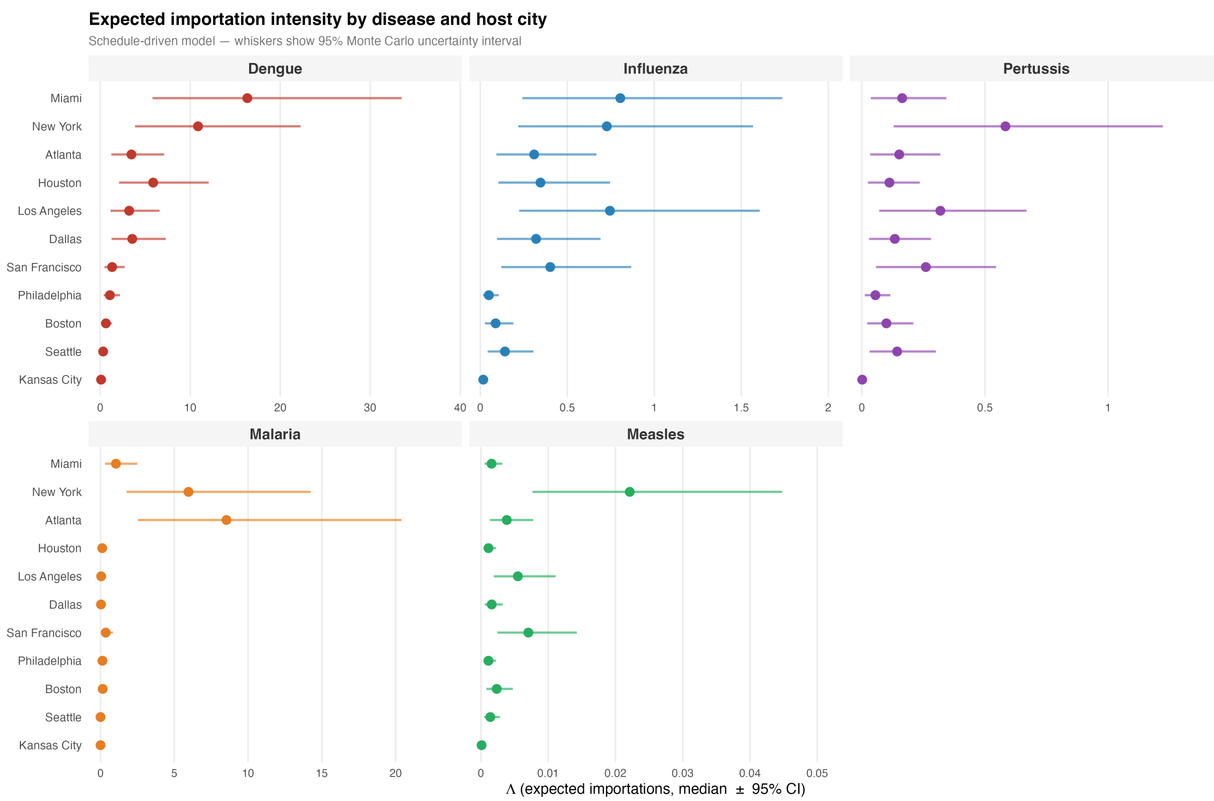


Figure 2: Expected importation intensity ($\Lambda$) with 95% Monte Carlo uncertainty intervals under the schedule-driven model (Model 3). Each panel shows one disease; cities are ordered from highest to lowest median $\Lambda$. Points show the MC median; horizontal whiskers span the 95% interval (5,000 Uniform draws on $\rho_d$ and $p_d$). Note free x-axis scales: dengue intensity is orders of magnitude above the other diseases.

The city and disease rankings visible in Figure 2 are robust across all three model tiers (Figure 8 in Appendix B). The step from the Baseline to the WC-adjusted model produces a near-uniform ∼17% uplift in $\Lambda$ ($\phi_{\text{WC}} = 1.174$; the WC fan increment accounts for approximately 8.3% of projected arrivals for qualified nations); the further step to the schedule-driven model introduces modest city-specific divergence without altering rankings. Background tourism—travel that would occur regardless of the tournament—accounts for the dominant share of importation risk even for strongly WC-associated nations such as

Brazil and Colombia; non-qualified countries contribute exclusively through this background stream.

## Source country drivers

Figure 3 shows the top 10 source countries by total expected importations for each disease, coloured by world region, with 95% Monte Carlo uncertainty intervals. Country rankings are robust: the top three contributors per disease are stable across the full parameter range. The geographic pattern is disease-specific. Latin America—Brazil, Colombia, Venezuela—dominates dengue and influenza, reflecting both high endemic burden and the largest travel volumes to the US among WC-qualified nations. West and Central Africa (Nigeria, Ghana, Senegal, Zaire/DRC) drives malaria risk. European nations (Germany, the United Kingdom, France) lead pertussis and measles, consistent with the documented pertussis resurgence and ongoing measles circulation in Europe. Brazil is the single highest-risk source nation overall, appearing in the top contributors for dengue, malaria, influenza, and pertussis simultaneously.

# Extensions and Generalizability

Although applied here to the 2026 FIFA World Cup, the Poisson importation framework is deliberately general. The three core inputs—a travel volume matrix, a country-level disease incidence table, and a travel-while-infectious probability—can be assembled for any disease, any set of destination cities, and any international gathering. The WC2026 application is a case study, not a constraint.

**Additional diseases.** The model applies to any pathogen for which country-level incidence and a travel-while-infectious probability can be specified. The current analysis covers five diseases (dengue, influenza, malaria, measles, pertussis); two near-term extensions are *chikungunya* and *mumps*. Chikungunya poses particularly high importation risk given the 2025–2026 outbreak in Brazil and Argentina (concurrent analyses estimate >10 expected monthly imports from Brazil alone at baseline [12]); monthly case data are available from PAHO and ECDC. Mumps data are available from the same WHO Immunization portal used for measles and pertussis. Mpox (Clade I) and rubella are lower-priority additions given low importation probabilities under current outbreak conditions. For any newly emerging pathogen, the framework requires only a case series by country and an estimate of the travel-while-infectious window.

**Beyond venue cities.** The current analysis covers the 11 US venue cities represented

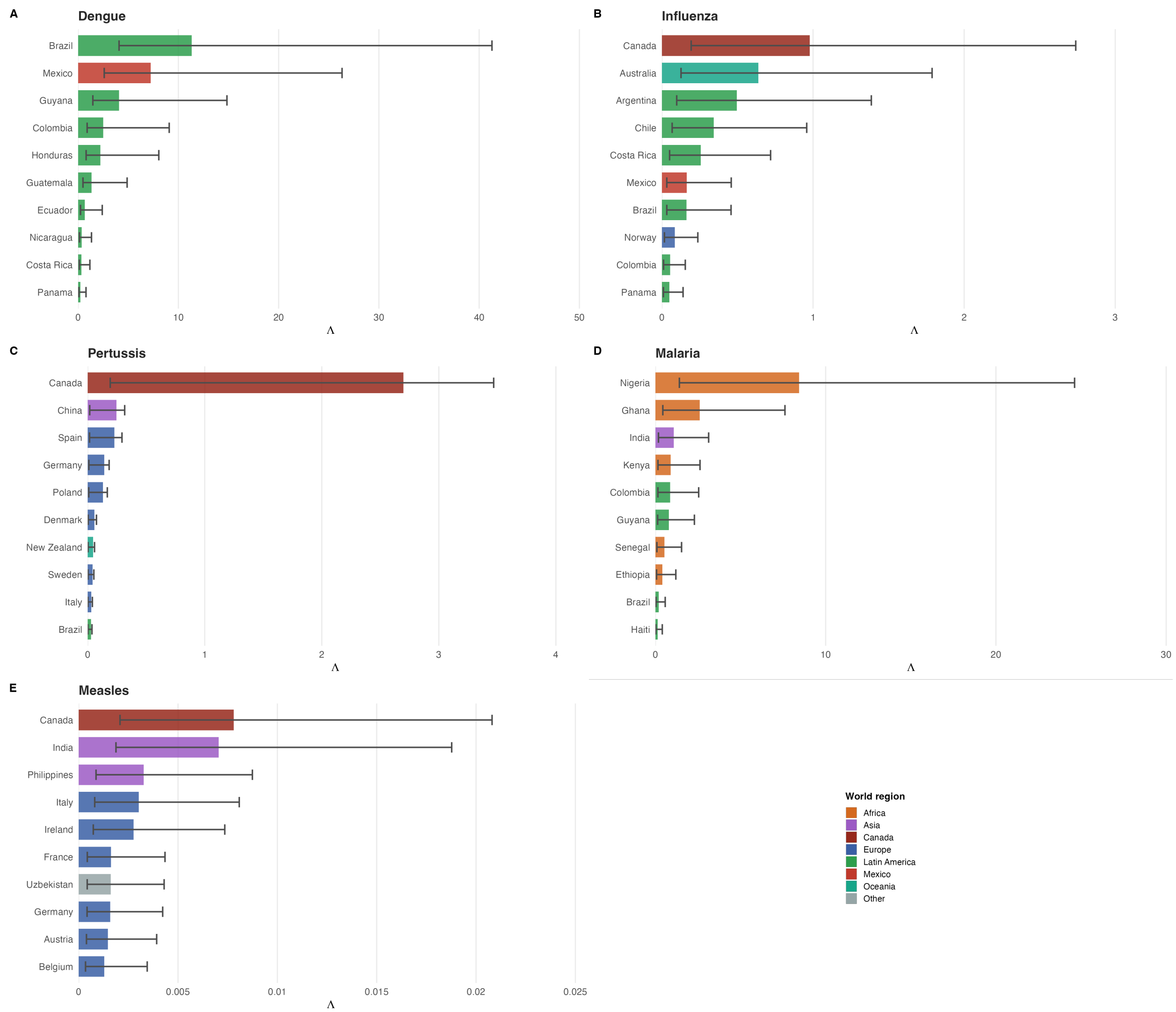


Figure 3: Top 10 source countries by total expected importations (schedule-driven model, summed across all US venue cities) for each disease. Bars show the MC median; whiskers show the 95% uncertainty interval. Colors indicate world region. Latin America dominates dengue and influenza risk; Africa drives malaria; Europe is the primary source for pertussis and measles.

in BTS T-100 routing data. The framework extends immediately to any US city or county by substituting an alternative routing source—DOT enplanement data, commercial booking records, or domestic connection data—for the T-100 fraction $f_{c,v}$. This is directly relevant for cities connected to venue cities by domestic flights, where internationally arriving travelers may complete their journeys on domestic legs not captured by T-100. The same logic applies beyond the United States: any country with international arrival records and disease surveillance data can be modeled as a destination.

**Other mass gatherings and routine surveillance.** The three-model hierarchy applies to any large international gathering: Olympic Games, religious pilgrimages (e.g., Hajj), international summits, or music festivals. The only event-specific inputs are the growth factor decomposition and the venue routing schedule; all other components transfer directly. Beyond mass gatherings, Model 1 functions as a standing importation surveillance tool: updated monthly with current COR arrivals and WHO disease data, it provides a near-real-time estimate of importation pressure at any US airport city, independent of any scheduled event.

# Discussion

The dominant empirical finding of this analysis—that background tourism accounts for the majority of importation risk even during a major international sporting event—carries a direct practical implication: public health preparedness should not focus exclusively on World Cup fans but should treat the entire June international visitor population as the primary exposure group.

The uncertainty analysis reveals important asymmetries in the reliability of estimates across diseases. Dengue importation risk is robust across the full parameter space, remaining near-certain at the highest-risk cities regardless of detection-rate assumptions. Pertussis estimates, by contrast, should be interpreted as conservative upper bounds: the central detection rate ($\rho = 0.10$) lies at the upper end of the literature range, so true importation intensity is more likely to fall below the median than above it. More broadly, diseases whose central detection rates fall near the lower bound of the literature range (dengue, influenza) may be systematically underestimated, a finding with direct implications for surveillance prioritization.

The stability of city rankings across all three model tiers—with New York, Los Angeles, Miami, and Houston consistently at highest aggregate risk—demonstrates that the principal findings are robust to uncertainty in WC-specific travel projections. A baseline analysis using only routine June 2024 arrivals (Model 1) identifies the same priority cities as the schedule-driven model (Model 3), supporting the practical utility of the framework even when event-specific inputs are unavailable.

Quantifying influenza alongside vector-borne and vaccine-preventable diseases reflects an important seasonal consideration: June coincides with the Southern Hemisphere winter influenza peak, and WC-qualified nations in South America and Oceania carry substantial seasonal burden during the tournament window.

This framework is designed for transparency and reproducibility: all data sources are publicly available and all code is openly shared, enabling any public health team to deploy, update, and adapt the analysis without proprietary data agreements. This positions the framework as complementary to concurrent analyses employing airline booking records and commercial mobility models [12], which are suited to different institutional contexts. The WC2026 estimates are broadly consistent with those of more resource-intensive models for endemically stable diseases (dengue, malaria), validating the public-data approach for first-order risk assessment. For outbreak-sensitive diseases, discrepancies arising from the use of annual rather than rolling surveillance data—most evident for UK measles and Norwegian pertussis—quantify the premium on data currency for rapidly evolving pathogens and motivate real-time data linkage as a priority methodological improvement.

The three-model structure maps onto a practical planning timeline: Model 1 can be executed as soon as host cities are announced; Model 2 updated once attendance projections become available; and Model 3 finalised once the match schedule is confirmed. City-specific estimates and country-level contributions can inform decisions on which venues and source-country streams to prioritise for enhanced surveillance, though operational decisions require additional local context—vector presence, population immunity, and healthcare capacity—not captured here. The Monte Carlo uncertainty intervals allow decision-makers to distinguish robust findings from estimates warranting caution, supporting more calibrated risk communication than a single point estimate permits. Beyond event-specific applications, the simplicity of the data requirements opens the possibility of a continuous monitoring mode: updated monthly with current COR arrivals and WHO surveillance figures, the framework can track importation pressure at US international gateways as a standing epidemiological tool, independent of any scheduled mass gathering.

Several key limitations bound the scope of these estimates; full detail is given in Appendix C. BTS T-100 data cover nonstop international segments only; passengers connecting through foreign hubs lose their country identity, and passengers connecting through US hubs are assigned to the gateway city rather than their final destination. Disease incidence is derived from annual WHO reporting cycles, which underestimate risk during active outbreaks (UK measles, European pertussis surges) and may reflect prior-year dynamics rather than the current transmission season. Growth factors $\phi_c$ are aggregate projections; actual June 2026 travel volumes will differ. The model estimates importation risk only—it does not model onward local transmission, which depends on vector presence, population immunity, and healthcare capacity not captured here. Finally, the Monte Carlo analysis propagates uncertainty in disease parameters ($\rho_d$, $p_d$) but treats travel volumes as fixed; uncertainty in the NTTO projections, T-100 routing fractions, and the WC growth factor decomposition would widen all intervals further.

The highest-priority methodological improvement is timelier disease data: substituting

WHO FluNet-style rolling surveillance averages for annual incidence tables—where such data exist—would close the data-currency gap most evident for measles and pertussis. Extensions to additional diseases (chikungunya, mumps) and to domestic-connection cities beyond T-100 gateways are described in the Extensions and Generalizability section above. Longer term, coupling importation estimates with local vector surveillance and population immunity data would enable a full importation-to-transmission risk pathway, the primary gap between what this framework currently provides and what would be needed for outbreak probability estimation.

# Conclusions

We present a general, reproducible Poisson framework for quantifying infectious disease importation risk that requires only publicly available travel and surveillance data. The framework is applicable to any disease, any destination city with international arrivals, and any mass gathering or routine surveillance context. Applied to the 2026 FIFA World Cup across five diseases and 11 US venue cities, dengue fever emerges as the primary importation concern, with New York, Los Angeles, Miami, and Houston at highest aggregate risk. Influenza ranks second, reflecting its Southern Hemisphere winter peak during the June tournament window. Background international travel—not World Cup fan travel specifically—is the dominant risk driver for all diseases, a finding with direct implications for surveillance prioritization at any international event.

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# Appendix A: Detailed Methods and Data Sources

## Study setting

The 2026 FIFA World Cup will be hosted jointly by the United States, Mexico, and Canada, with matches scheduled from June 11 through July 19, 2026. The United States will host the majority of matches across eleven venues: MetLife Stadium (East Rutherford, NJ), SoFi Stadium (Inglewood, CA), AT&T Stadium (Arlington, TX), Mercedes-Benz Stadium (Atlanta, GA), NRG Stadium (Houston, TX), Arrowhead Stadium (Kansas City, MO), Lincoln Financial Field (Philadelphia, PA), Hard Rock Stadium (Miami Gardens, FL), Levi's Stadium (Santa Clara, CA), Lumen Field (Seattle, WA), and Gillette Stadium (Foxborough, MA). Canada hosts two venues (BC Place, Vancouver; BMO Field, Toronto) and Mexico hosts three (Estadio Azteca, Mexico City; Estadio BBVA, Monterrey; Estadio Akron, Guadalajara). The geographic distribution of all 16 venues across the three co-host nations is shown in Figure 4. All 48 qualified nations are listed in Supplementary Table 1.

We estimated the importation risk of five infectious diseases—dengue fever, influenza, malaria, measles, and pertussis—for each host city using three complementary models of increasing resolution: (1) a *baseline model* using COR June 2024 arrivals with BTS T-100 routing fractions and no World Cup growth adjustment; (2) a *World Cup (WC)-adjusted model* that scales volumes to country-level 2026 projections; and (3) a *schedule-driven model* that separates World Cup fan travel from background tourism and routes fans to the specific venue cities where their national team plays. The five diseases were selected because each is endemic in countries with strong football traditions, each has a plausible pathway for travel-associated importation into the United States, and each represents a distinct surveillance and public health response challenge. Influenza is included because June corresponds to the Southern Hemisphere winter peak, coinciding with the tournament window for WC-qualified nations in South America and Oceania.

## Notation and variable definitions

The following indices and variables are used consistently throughout all three models. Where the range of an index differs between models, this is noted explicitly.

### Indices

- $c$ — source **country** of international travelers (all three models).
- $v$ — **destination venue city**. In all three models, $v$ ranges over the eleven US WC venue cities (Table 2). Model 3 additionally covers the five non-US host cities (Toronto,

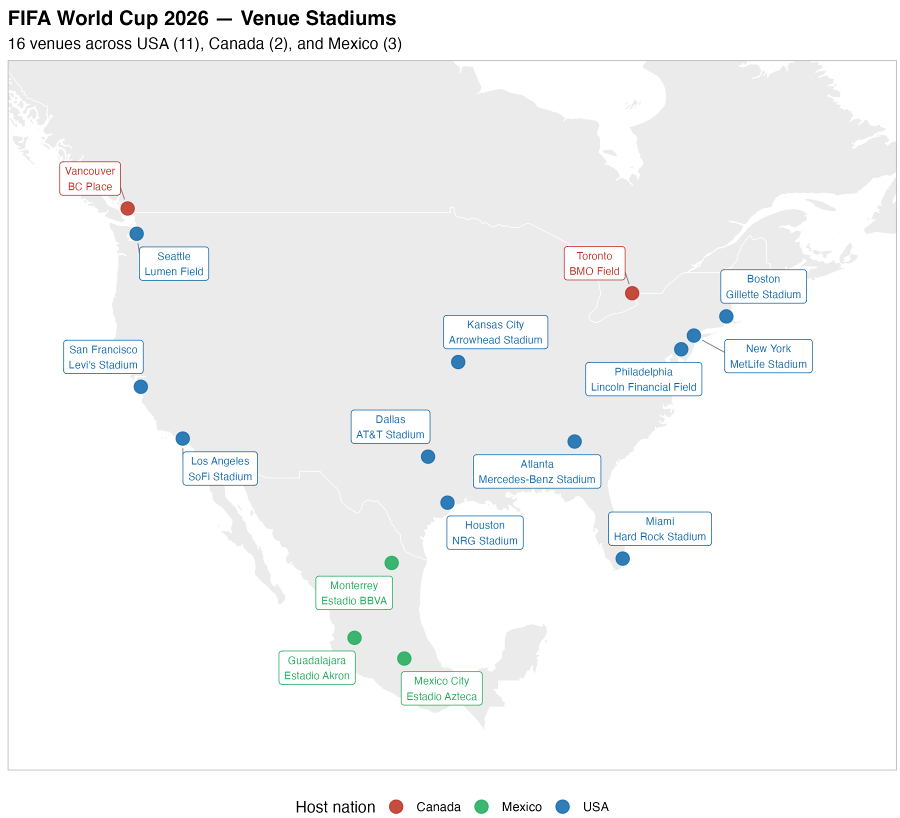



Figure 4: Geographic distribution of the 16 FIFA World Cup 2026 venue stadiums across the three co-host nations. Circles are colour-coded by host country: USA (blue, 11 venues), Canada (red, 2 venues), and Mexico (green, 3 venues). Suburb stadium locations are labelled by their metropolitan area to match the spatial aggregation used throughout the analysis (e.g. East Rutherford → New York, Foxborough → Boston).

Vancouver, Guadalajara, Mexico City, Monterrey).

- $d$ — **disease**: $d \in$ {dengue, influenza, malaria, measles, pertussis}.
- $y$ — **calendar year**, used when averaging over multiple June cross-sections ($y \in \{2023, 2024, 2025\}$ for T-100 routing fractions).

### Travel and importation variables

- $N_{c,v}$ — expected number of **travelers** from source country $c$ arriving in city $v$ during June. Constructed differently in each model (see below).
- $\Lambda_{v,d}$ — **importation intensity** (Poisson rate): expected number of imported cases of disease $d$ arriving in city $v$ during June 2026.

- $\lambda_{c,v,d}$ — **importation contribution** from source country $c$ to city $v$ for disease $d$; $\Lambda_{v,d} = \sum_c \lambda_{c,v,d}$.

### Disease and parameter variables

- $I_{c,d}$ — **per-traveler infection probability**: probability that a randomly selected traveler from country $c$ is carrying disease $d$ during their trip. Defined as $I_{c,d} = \rho_d \cdot C_{c,d}/P_c$.
- $C_{c,d}$ — **reported cases** of disease $d$ in source country $c$ during June (or annual figure divided by 12 for diseases with only annual reporting).
- $P_c$ — **population** of source country $c$.
- $\rho_d$ — **reporting adjustment factor** for disease $d$: corrects for under-reporting in routine surveillance and the fraction of reported cases that are actively infectious at any given time (Table 3).
- $p_d$ — **travel-while-infectious probability**: probability that a person infected with disease $d$ continues to travel internationally during the infectious window (Table 3).

### Routing fractions and travel growth

- $f^{\mathrm{T100}}_{c,v}$ — **T-100 routing fraction**: share of country $c$'s US-bound June arrivals that enter through city $v$, derived from BTS T-100 International Segment data averaged over June 2023–2025. Used in all three models.
- $\phi_c$ — **growth factor** for country $c$: ratio of projected 2026 to observed 2024 annual visitor volume ($\phi_c = V_{c,2026}/V_{c,2024}$). Derived from NTTO for Tier 1 countries; set to $\phi_{\mathrm{WC}}$ or $\phi_{\mathrm{base}}$ for all others.
- $\phi_{\mathrm{WC}} = 1.174$ — aggregate WC growth factor (NTTO total: 85,017/72,390 thousand).
- $\phi_{\mathrm{base}} = 1.077$ — baseline (non-WC) growth factor derived as the geometric mean annual growth rate over June 2023–2025, averaged across COR (1.062) and T-100 (1.092) data sources; applied to non-qualified countries.
- $N^{\mathrm{WC}}_c$ — **WC-fan travel increment**: marginal arrivals attributed to the World Cup for country $c$.
- $N^{\mathrm{bg}}_c$ — **background arrivals**: June 2026 travel that would have occurred without the WC.

- $\omega_{c,v}$ — **schedule routing fraction**: share of WC fans from country $c$ attending matches in venue city $v$; defined as games played in $v$ divided by total known games for that team.

- $\mathcal{Q}$ — set of WC-qualified countries with a known group-stage match schedule (TBD matches excluded).

## Importation risk framework

We modelled disease importation as a Poisson process. This choice is standard for rare-event counting problems: when individual import events are independent and the per-traveller probability of infection is small, the number of imported cases in a fixed time window converges to a Poisson random variable. The Poisson framework is also analytically tractable and additive over source units, allowing country-level and stream-level contributions to be computed and summed independently.

If $\Lambda_{v,d}$ is the expected number of imported cases of disease $d$ arriving in city $v$ during June 2026, the probability of observing at least one importation is:

$$P(X_{v,d} \geq 1) = 1 - e^{-\Lambda_{v,d}} \tag{2}$$

where $\Lambda_{v,d} = \sum_c \lambda_{c,v,d}$ sums over all origin countries $c$. This framework is consistent with previously published importation risk analyses for large international sporting events [13].

Each country-level contribution takes the same multiplicative form across all three models:

$$\lambda_{c,v,d} = N_{c,v} \cdot I_{c,d} \cdot p_d \tag{3}$$

where $N_{c,v}$ is the number of travellers from country $c$ arriving in city $v$, $I_{c,d}$ is the per-traveller probability that an arriving person carries disease $d$, and $p_d$ is the probability that such a person travels internationally while infectious. The three models differ in how $N_{c,v}$ and $I_{c,d}$ are constructed.

## Baseline model: COR June 2024 travel volume with T-100 routing (no WC adjustment)

### Rationale and design

The baseline model establishes the importation risk expected in a typical June *without* any World Cup effect. Its purpose is to provide a pre-tournament reference against which the

WC-driven changes in Models 2 and 3 can be measured. Three design choices ensure this reference is both accurate and spatially consistent:

1. **Travel volume: COR June 2024, no growth factor.** Monthly air arrivals by country of residence (COR/I-94) for June 2024 are used directly, without any WC growth factor ($\phi_c = 1$ for all countries). June 2024 is the most recent pre-tournament June with complete data and represents normal summer travel patterns.

2. **City routing: BTS T-100 routing fractions.** Country-level routing fractions $f_{c,v}^{\text{T100}}$ (described in the next subsection and in Equation 8) allocate each country's COR arrivals to the eleven US venue cities. Using T-100 from the baseline ensures all three models operate on the same spatial grid, making the three-way comparison interpretable: the step from Model 1 to Model 2 isolates the WC travel surge alone, not a change in routing methodology.

3. **Disease incidence: country-level.** Country-specific incidence proxies $I_{c,d}$ are used (Equation 12), providing the same precision as Models 2 and 3 and ensuring that differences between models reflect travel volume changes, not incidence aggregation.

## Baseline city-level arrivals

Baseline arrivals from country $c$ to venue city $v$ are:

$$N_{c,v}^{\text{baseline}} = N_{c,\text{June 2024}}^{\text{COR}} \cdot f_{c,v}^{\text{T100}} \tag{4}$$

Countries with no T-100 routing data (typically nations with no scheduled nonstop international service to the United States) are excluded; their volumes are negligible in the aggregate.

## Importation intensity (baseline)

$$\lambda_{c,v,d} = N_{c,v}^{\text{baseline}} \cdot I_{c,d} \cdot p_d \tag{5}$$

$$\Lambda_{v,d} = \sum_{c} \lambda_{c,v,d} \tag{6}$$

The probability of at least one importation is computed via Equation 2.

# BTS T-100 International Segment data: country-level city routing fractions

## Motivation and data source

All three models require country-level city routing fractions that cover every US WC venue city. We use the Bureau of Transportation Statistics (BTS) T-100 International Segment (All Carriers) dataset [3], which records nonstop international flight segments landing at US airports at the individual airport level. T-100 covers all US commercial airports and reports passengers by country of origin (via the departure country code), making it possible to compute country-specific routing fractions for all eleven US venue cities. The T-100 data do not include an origin city field, only the originating country code for the international segment.

## Data processing and routing fraction construction

Three full years of T-100 data were downloaded for 2023, 2024, and 2025, covering all months. Only June records (month = 6) were retained for each year, yielding three June cross-sections. This window was chosen to: (1) match the June travel period of the WC group stage; (2) use the three most recent complete pre-tournament June months; and (3) avoid the COVID-recovery period (2020–2022) during which route networks and traffic volumes were atypical.

Records were filtered to scheduled passenger service only (`CLASS = "F"`), excluding charter (`N`), all-cargo (`L`), and mixed cargo (`P`) service classes. Only segments with at least one passenger (`PASSENGERS > 0`) and landing in the United States (`DEST_COUNTRY = "US"`) were retained. Destination airports were then mapped to the eleven US WC venue city groups using a predefined airport-to-city table (e.g., JFK, EWR, and LGA → New York; MIA and FLL → Miami; SFO, OAK, and SJC → San Francisco). Any airport not associated with a WC venue city was excluded.

For each source country $c$, venue city $v$, and year $y$, the routing fraction is first computed within each June cross-section as:

$$f^{\mathrm{T100}}_{c,v,y} = \frac{N^{\mathrm{June}}_{c,h,y}}{N^{\mathrm{June}}_{c,\mathrm{US},y}} \tag{7}$$

where $N^{\mathrm{June}}_{c,h,y}$ is the total T-100 passengers from country $c$ to the airport group serving venue city $v$ in June of year $y$, and $N^{\mathrm{June}}_{c,\mathrm{US},y}$ is the total T-100 passengers from country $c$ to *all* US airports in June of year $y$ (the denominator). Using the full US total—not just WC venue airports—ensures that fractions are true proportions, interpretable as the share of US-bound travel from country $c$ that enters through each city.

The mean routing fraction is then averaged across the three years:

$$f_{c,v}^{\mathrm{T100}} = \frac{1}{|Y|} \sum_{y \in Y} f_{c,v,y}^{\mathrm{T100}} \tag{8}$$

where $Y = \{2023, 2024, 2025\}$. Averaging within years before averaging across years (rather than pooling all passenger counts) prevents high-volume years from dominating the estimate and keeps each annual fraction a valid proportion.

### Important limitations of T-100 routing fractions

T-100 records the *nonstop international segment* only. A passenger travelling on a connecting itinerary (e.g., São Paulo → Miami → Kansas City) appears only in the Miami row, because Miami is the airport where the international segment terminates and US Customs is cleared. Kansas City, as a domestic connecting leg, is not recorded in T-100 for that passenger. This is the correct interpretation for importation risk modelling: from a public health perspective, the traveller enters the US in Miami, where surveillance and border health screening occurs. Kansas City correctly shows near-zero T-100 routing fractions for most countries, reflecting minimal direct international service; its WC-specific traffic is captured entirely by the schedule-driven WC-fan stream.

The T-100 data cover US airports only. Routing fractions for the five non-US host cities (Toronto, Vancouver, Guadalajara, Mexico City, Monterrey) cannot be derived from T-100 and are not estimated here; those cities receive importation risk estimates solely from the WC-fan travel stream.

Country names in T-100 use ISO/ICAO conventions (e.g., "Korea, Republic of", "Viet Nam") while the CBP COR arrival data use informal short names (e.g., "South Korea", "Vietnam"). A standardised recode table covering 26 known mismatches was applied before joining the two datasets.

## World Cup–adjusted model: country-level travel volume (2026 projections)

### Motivation

The baseline model uses June 2024 COR arrivals without any World Cup adjustment, capturing normal summer travel patterns. It does not account for the additional influx of visitors attending the World Cup. The WC-adjusted model addresses this limitation by projecting June 2026 arrival volumes at the country level using NTTO forecasts and COR data scaled by WC growth factors. All other modelling choices (T-100 routing fractions, country-level disease incidence, same $\rho_d$ and $p_d$ values) are unchanged, so the Model 1 to Model 2 comparison isolates the effect of the WC travel surge alone.

### Construction of 2026 travel volume

Country-level June 2026 visitor volumes to the United States were constructed from two complementary sources using a tiered approach. The tiered design is necessary because not all countries have official NTTO projections; for the majority of the world, the only available country-level arrival data are the CBP COR monthly figures, which are then scaled by an appropriate growth factor.

**Tier 1 — NTTO projections (top 12 source markets).** Official 2026 international visitor projections for twelve major source markets were obtained from the US National Travel and Tourism Office (NTTO) 2025 Forecast Report [2]. These projections—available for Australia, Brazil, Canada, China, France, Germany, India, Italy, Japan, Mexico, South Korea, and the United Kingdom—already incorporate the World Cup tourism effect and should *not* have additional WC visitors added on top. The 2024-to-2026 growth factor for each country was derived as $\phi_c = V_{c,2026}/V_{c,2024}$, where $V_{c,y}$ is the projected annual visitor volume in year $y$ (Table 4). Using the NTTO projections for these twelve countries is important because they reflect expert forecasts that account for the specific WC effect, bilateral travel trends, and macroeconomic factors, making them more reliable than a simple scaling of 2024 COR arrivals.

**Tier 2 — COR data scaled by global growth factors (remaining countries).** For all other countries, country-level June 2024 arrivals were extracted from the US Customs and Border Protection (CBP) I-94 Monthly Arrivals by Country of Residence dataset [1]. The COR data provide monthly arrival counts by country of residence and are the most granular publicly available source of international arrival data for the United States. A growth factor was then assigned based on World Cup qualification status:

- *WC-qualified countries not in Tier 1*: scaled by the overall NTTO 2024→2026 growth factor ($\phi_{\mathrm{WC}}$ = 85,017/72,390 = 1.174). This factor was derived from the aggregate NTTO total and incorporates the WC tourism uplift at the global level, providing a reasonable approximation for WC-qualified countries not individually projected by NTTO.

- *Non-qualified countries*: scaled by a data-driven baseline growth factor $\phi_{\text{base}}$ estimated from observed trends in US inbound travel independent of the tournament. We derived $\phi_{\text{base}}$ from two sources covering June 2023–2025: (i) CBP COR monthly arrivals and (ii) BTS T-100 total international passengers arriving in the US. For each source, the geometric mean annual growth rate over the two observed intervals is:

$$\bar{g} = \left(\frac{N_{\text{June,2025}}}{N_{\text{June,2023}}}\right)^{1/2} = \sqrt{\frac{N_{2024}}{N_{2023}} \cdot \frac{N_{2025}}{N_{2024}}} \tag{9}$$

  and the two-year forward factor follows as $\phi_{\text{base}} = \bar{g}^2 = N_{\text{June,2025}}/N_{\text{June,2023}}$. The COR estimate is $\phi_{\text{base}}^{\text{COR}} = 1.062$ (June totals: 4,970K in 2023, 5,279K in 2025) and the T-100 estimate is $\phi_{\text{base}}^{\text{T100}} = 1.092$ (passengers: 10,700K in 2023, 11,686K in 2025). The two sources agree closely; their arithmetic mean, $\phi_{\text{base}} = 1.077$, is used in the model. This value is lower than a simple extrapolation of pre-2024 growth because June 2025 arrivals declined relative to June 2024 in both datasets, reflecting a plateau after the post-COVID travel rebound rather than continued acceleration.

The projected June 2026 arrivals for country $c$ are thus:

$$N_{c,\text{June 2026}} = N_{c,\text{June 2024}}^{\text{COR}} \cdot \phi_c \tag{10}$$

**Country-to-city allocation via T-100 routing fractions.** City-level arrival volumes for country $c$ and venue city $v$ were estimated by applying the T-100 country-level routing fractions (Equation 8):

$$N_{c,v}^{2026} = N_{c,\text{June 2026}} \cdot f_{c,v}^{\text{T100}} \tag{11}$$

This uses the same T-100 routing fractions as the baseline model (Equation 4), but applied to the projected 2026 volumes. Countries with no T-100 routing data are excluded from the city-level allocation; their volumes are negligible.

## Country-level disease incidence

Disease burden is computed at the country level across all three model tiers, providing fine-grained precision that reflects within-region heterogeneity in disease burden. For example, Brazil and Argentina differ substantially in dengue incidence despite belonging to the same broad regional grouping. Country-level incidence is defined as:

$$I_{c,d} = \rho_d \cdot \frac{C_{c,d}}{P_c} \tag{12}$$

where $C_{c,d}$ is the reported cases of disease $d$ in country $c$ and $P_c$ is the national population (2020 estimates from the UN World Population Prospects [14]). The same $\rho_d$ and $p_d$ values are used in all three models to ensure that model comparisons reflect travel volume differences alone.

### Importation intensity (WC-adjusted)

$$\lambda_{c,v,d} = N_{c,v}^{2026} \cdot I_{c,d} \cdot p_d \tag{13}$$

$$\Lambda_{v,d} = \sum_{c} \lambda_{c,v,d} \tag{14}$$

The probability of at least one importation is then computed via Equation 2.

## Schedule-driven venue routing model

### Motivation

Both the baseline and WC-adjusted models distribute all international arrivals across venue cities using T-100 routing fractions, which reflect habitual tourist flow patterns. This is appropriate for background tourism but does not capture the WC-specific fan travel logic: a supporter from Brazil attending a group-stage match in Houston is unlikely to distribute to Boston, Philadelphia, or Seattle in the same proportions as a regular tourist. The schedule-driven model captures this distinction by decomposing total travel into a WC-specific fan stream—routed by the match schedule—and a background stream routed by T-100 fractions.

### Travel decomposition

For each source country $c$, total projected June 2026 arrivals $N_{c,\text{June 2026}}$ (Equation 10) are decomposed into two components:

- *WC-fan stream*: the marginal arrivals attributable to the World Cup, defined as the increment above the no-WC counterfactual (the volume that would have arrived with only baseline growth):

$$N_c^{\text{WC}} = N_{c,\text{June 2024}}^{\text{COR}} \cdot \max\bigl(0,\ \phi_c - \phi_{\text{base}}\bigr) \tag{15}$$

  This formulation isolates the WC-specific increment: any growth above the baseline rate ($\phi_{\text{base}} = 1.077$) is attributed to the tournament.

- *Background stream*: arrivals that would have occurred regardless of the World Cup:

$$N_c^{\text{bg}} = N_{c,\text{June 2026}} - N_c^{\text{WC}} \tag{16}$$

For countries with NTTO growth factors below the baseline ($\phi_c < \phi_{\text{base}} = 1.077$), the WC increment is set to zero and all travel is treated as background. For a WC-qualified country assigned the global WC growth factor ($\phi_{\text{WC}} = 1.174$), the WC fan fraction is approximately 8.3% of projected arrivals: $(1.174 - 1.077)/1.174 \approx 0.083$.

## Schedule-based WC fan routing

The 2026 FIFA World Cup match schedule was extracted from official FIFA event records for all 48 qualifying nations [4]. Matches with at least one undetermined opponent (labelled "TBD" in the schedule) are excluded from the fan routing calculation; both group-stage and knockout-round entries with TBD opponents are dropped. For each WC-qualified country $c$ with team $t_c$, a schedule routing fraction is defined as:

$$\omega_{c,v} = \frac{g_{c,v}}{G_c} \tag{17}$$

where $g_{c,v}$ is the number of matches played by team $t_c$ at venue city $v$ and $G_c = \sum_v g_{c,v}$ is the total number of known matches for that team. This fraction represents the share of WC fans from country $c$ expected to be present in venue city $v$, under the assumption that fan travel is proportional to the number of matches played there. WC-fan arrivals at each venue city are then:

$$N_{c,v}^{\text{WC}} = N_c^{\text{WC}} \cdot \omega_{c,v} \tag{18}$$

## Background stream routing via T-100 fractions

Background travellers (Equation 16) are assumed to follow normal routing patterns, independent of the match schedule. We use the T-100 country-level routing fractions (Equation 8) to distribute background arrivals across venue cities:

$$N_{c,v}^{\text{bg}} = N_c^{\text{bg}} \cdot f_{c,v}^{\text{T100}} \tag{19}$$

This is consistent with the routing approach used in the baseline and WC-adjusted models. Because T-100 covers all 11 US venue cities, background travelers contribute to importation risk across the full spatial grid. Countries with no direct T-100 service to a given city have a zero routing fraction for that city, correctly reflecting minimal direct international service there.

**Note on multi-team countries.** England and Scotland are distinct World Cup participants but share a single country-of-residence entry ("United Kingdom") in the COR arrivals data. Both teams' match venues are combined under this single entry, and the routing fractions $\omega_{\text{UK},v}$ are computed over the union of their games. Because the fractions sum to one, no double-counting of UK arrivals occurs.

**Venue cities.** The schedule-driven model covers all sixteen host venues across the three co-host nations (Table 2). For all eleven US venue cities, both the WC-fan and background streams contribute to importation risk: the WC-fan stream routes fans via the match schedule ($\omega_{c,v}$) and the background stream routes tourists via T-100 fractions ($f_{c,v}^{\text{T100}}$). For the five non-US host cities (Toronto and Vancouver in Canada; Guadalajara, Mexico City, and Monterrey in Mexico), T-100 fractions are not available and only the WC-fan stream contributes; background tourist flows to those cities are not estimated.

Table 2: WC 2026 host venues and their canonical city names used in the schedule-driven model. Cities with T-100 background routing are marked with †.

| **Stadium** | **City (canonical)** | **Country** | **T-100 bg?** |
|---|---|---|---|
| MetLife Stadium | New York† | USA | Yes |
| SoFi Stadium | Los Angeles† | USA | Yes |
| AT&T Stadium | Dallas† | USA | Yes |
| Mercedes-Benz Stadium | Atlanta† | USA | Yes |
| NRG Stadium | Houston† | USA | Yes |
| Arrowhead Stadium | Kansas City† | USA | Yes* |
| Lincoln Financial Field | Philadelphia† | USA | Yes |
| Hard Rock Stadium | Miami† | USA | Yes |
| Levi's Stadium | San Francisco† | USA | Yes |
| Lumen Field | Seattle† | USA | Yes |
| Gillette Stadium | Boston† | USA | Yes |
| BC Place | Vancouver | Canada | No |
| BMO Field | Toronto | Canada | No |
| Estadio Azteca | Mexico City | Mexico | No |
| Estadio BBVA | Monterrey | Mexico | No |
| Estadio Akron | Guadalajara | Mexico | No |

* Kansas City shows near-zero T-100 fractions for most countries, reflecting its minimal direct international air service; its WC-specific traffic is captured by the WC-fan stream.

### Importation intensity (schedule-driven)

The total Poisson importation intensity for venue city $v$ is the sum of contributions from both travel streams across all source countries:

$$\Lambda_{v,d} = \underbrace{\sum_{c \in \mathcal{Q}} N_{c,v}^{\mathrm{WC}} \cdot I_{c,d} \cdot p_d}_{\text{WC-fan stream}} + \underbrace{\sum_{c} N_{c,v}^{\mathrm{bg}} \cdot I_{c,d} \cdot p_d}_{\text{background stream}} \tag{20}$$

where $\mathcal{Q}$ denotes the set of WC-qualified countries with a known group-stage schedule. The probability of at least one importation is computed via Equation 2.

## Three-model comparison

To quantify the effect of each modelling assumption, results from the three models are compared side by side for each disease and destination city. The three models form a nested hierarchy:

1. **Baseline**: COR June 2024 arrivals (no WC adjustment), country-level disease incidence, all eleven US venue cities via T-100 routing fractions. Establishes the pre-tournament reference.

2. **WC-adjusted**: Country-level June 2026 projections (NTTO + COR scaling by $\phi_c$), same country-level disease incidence, same T-100 routing, all eleven US venue cities.

3. **Schedule-driven**: Same 2026 projections decomposed into WC-fan and background streams; WC fans routed to all sixteen venue cities via the match schedule ($\omega_{c,v}$); background travelers routed via T-100 fractions to all eleven US cities.

The step from Model 1 to Model 2 isolates the effect of the WC travel surge (growth factors $\phi_c$ applied to 2024 base volumes). All other modelling choices—routing fractions, disease incidence, $\rho_d$, $p_d$—are held fixed. The step from Model 2 to Model 3 isolates the effect of schedule-based venue routing: WC fans are no longer spread by habitual tourist flows but directed to the specific cities where their team plays. The five non-US host cities (Toronto, Vancouver, Guadalajara, Mexico City, Monterrey) appear only in Model 3.

Results across all three model tiers are compared in Figure 8 (the dodged-bar chart in the main text), which compares $\Lambda$ for the six highest-risk cities across the three model tiers for each disease. Two summary metrics are used in the analysis:

- $P(\geq 1)$: probability of at least one importation (Equation 2). Useful as a communication metric but saturates quickly toward 1 for high-risk cities.

- $\Lambda$: expected number of imported cases (the raw Poisson intensity). Remains on a linear scale even when $P(\geq 1) \to 1$ and allows direct quantitative comparison across cities and models.

## Country-level importation contributions

The additive structure of the Poisson model allows city-level importation risk to be attributed to individual source countries. Total importation intensity decomposes as:

$$\Lambda_{v,d} = \sum_{c} \lambda_{c,v,d} \tag{21}$$

where $\lambda_{c,v,d} = N_{c,v} \cdot I_{c,d} \cdot p_d$ is the contribution from source country $c$ to destination city $v$ for disease $d$. Using the schedule-driven model (which separates WC-fan and background streams), three summaries are produced:

1. **Country importation ranking**: the top ten source countries by total expected importations (summed across all destination cities, $\sum_v \lambda_{c,v,d}$) are shown for each disease with 95% Monte Carlo uncertainty intervals (Figure 3 in the main text).

2. **Travel-stream decomposition**: for the top 20 countries by importation burden, the fraction of $\lambda_{c,v,d}$ attributable to the WC-fan stream versus the background stream is shown separately for each disease (Figure 10). Countries with WC teams are expected to show a non-zero WC-fan component; non-qualified countries will show only background travel contributions.

3. **Country × city importation heatmap**: a matrix visualisation of $\lambda_{c,v,d}$ summed over both streams, with rows ordered by total importation burden and columns by destination city (Appendix Figures 11–15). This reveals the spatial concentration of risk: WC-qualified countries show importation concentrated in their specific match venues, while non-qualified countries show diffuse contributions spread via T-100 routing fractions.

## Disease data sources

### Dengue

Monthly country-level dengue case counts were obtained from the WHO Global Dengue Surveillance dataset [5]. Dengue is the primary focus of the WC-adjusted and schedule-driven models because it is highly seasonal (peaking in June in the Southern Hemisphere, where several WC-qualified nations are located), is endemic across much of Latin America, the Caribbean, and Southeast Asia, and has a substantial fraction of infections that are asymptomatic or mild enough to allow continued travel. For all three models, the mean June case count per country over 2024–2025 was used directly, divided by national population to obtain a per-capita incidence. Countries with no data were excluded.

### Malaria

Country-level malaria incidence rates for 2024 (cases per 1,000 population) were obtained from the WHO Global Malaria Programme National Unit Data [6]. Annual incidence was divided by 12 to obtain an approximate monthly rate. Malaria was included because several WC-qualified countries (particularly in sub-Saharan Africa and parts of Asia) have high endemic transmission, and even low traveller infection probabilities translate into meaningful importation risk given the large travel volumes from Latin America and Africa. Countries were recoded to match naming conventions in the travel datasets where necessary (e.g., "Democratic Republic of the Congo" → "Zaire (formerly DRC)").

### Measles and pertussis

Annual country-level incidence rates (cases per 1,000,000 population) for measles and pertussis were obtained from the WHO Immunization Data portal [7]. Annual incidence was divided by 12 to obtain an approximate monthly rate. Measles was included because of its resurgence in multiple regions due to declining vaccination coverage, and because it has a high infectiousness ($R_0 \approx 12$–$18$) that makes even a single importation event meaningful. Pertussis was included because its catarrhal phase closely resembles a common cold, making it likely that infected individuals continue to travel internationally; its $p_d$ value (Table 3) is accordingly the highest of the five diseases.

### Influenza

Weekly country-level positive specimen counts were obtained from the WHO FluNet / Global Influenza Surveillance and Response System (GISRS) [7]. FluNet reports laboratory-confirmed influenza A and B specimens submitted by national sentinel surveillance sites; the combined total (`INF_ALL`) is used here as a proxy for reported incidence, with the under-reporting correction $\rho_d$ accounting for the gap between confirmed specimens and true community burden. June corresponds to ISO weeks 22–26. Mean June positive specimens per country were computed across 2023–2025 to produce a stable baseline signal. Country names were harmonised to COR conventions before joining (26 known mismatches recoded, e.g., "United Kingdom of Great Britain and Northern Ireland" → "United Kingdom"). Countries with zero mean June specimens were excluded.

Influenza was included because June coincides with the peak of the Southern Hemisphere winter influenza season. WC-qualified nations in South America (Brazil, Argentina, Uruguay, Colombia, Ecuador, Peru, Bolivia, Paraguay, Chile, Venezuela) and Oceania (Australia, New Zealand) carry substantial seasonal influenza burden during the tournament window, making this the first importation risk analysis for the 2026 World Cup to quantify influenza alongside vector-borne and vaccine-preventable diseases.

## Model parameters

Table 3 summarises the disease-specific parameters used in all three models. The adjustment factor $\rho_d$ captures the combined effect of under-reporting in routine surveillance and the fraction of reported cases that represent actively infectious individuals at a given time. Values were assigned based on published burden studies: dengue and pertussis are the most severely under-reported diseases, with global detection fractions well below 20% in most endemic settings [8, 9, 11]; malaria detection at $\rho_d = 0.20$ is consistent with WHO World Malaria Report methodology [6]; and measles at $\rho_d = 0.60$ reflects the higher reporting rates typical of middle- and high-income source countries with mandatory notification [10]. The travel infectiousness probability $p_d$ reflects the likelihood that a person with the disease travels internationally while infectious, and was informed by the typical severity and duration of symptoms relative to the incubation period and infectious window.

## NTTO 2026 visitor projections for top 12 source markets

## Software

All analyses were performed in R (version 4.5.2). The main importation pipeline (`importationRisk_main_with_uncertainty.R`) uses `tidyverse` for data wrangling and visualisation, `readxl` for Excel data ingestion, `janitor` for column-name standardisation, `lubridate` for date operations, and `cowplot` for multi-panel figure composition. Monte Carlo uncertainty propagation (5,000 Uniform draws on $\rho_d$ and $p_d$) is integrated directly into the main pipeline via the vectorised `compute_mc_summary()` helper. A dedicated data preparation script (`t100_routing_prep.R`) processes BTS T-100 data and must be run once before the main pipeline to generate `Data/t100_routing_fractions.csv`. FluNet influenza data are downloaded automatically from the WHO FluNet API on first run and cached locally. Code and data are available at:

Table 3: Disease-specific model parameters. The $\rho_d$ literature range is derived from published burden and under-reporting studies; the adopted value falls within or at the conservative end of that range. All $p_d$ values are supported by travel medicine literature. Key references are given for $\rho_d$; $p_d$ values are based on biological plausibility and importation-model precedent.

| **Disease** | $\rho_d$ **range** | $\rho_d$ | $p_d$ | **Rationale for** $p_d$ | **Key ref.** ($\rho_d$) |
|---|---|---|---|---|---|
| Dengue | 0.06–0.26 | 0.10 | 0.50 | Mild viraemic phase; ∼40% viremic on arrival | [8, 9] |
| Influenza | 0.033–0.20 | 0.10 | 0.50 | Moderate illness; many travel during early illness; SH winter peak in June | [7] |
| Malaria | 0.10–0.35 | 0.20 | 0.30 | Severe illness; most febrile after return | [6] |
| Measles | 0.40–0.80 | 0.60 | 0.05 | Prostrating; ambulatory only in short pre-rash window | [10] |
| Pertussis | 0.01–0.10 | 0.10 | 0.70 | Catarrhal stage mimics common cold; unrecognised | [11] |

Table 4: NTTO 2026 international visitor projections and derived growth factors used in the WC-adjusted model. Source: US National Travel and Tourism Office Forecast Report (March 2025).

| **Country** | **2024 (000s)** | **2026 (000s)** | $\phi_c$ | **WC team?** |
|---|---:|---:|---:|:---:|
| Canada | 20,241 | 23,015 | 1.137 | Yes (host) |
| Mexico | 16,990 | 19,557 | 1.151 | Yes (host) |
| United Kingdom | 4,037 | 4,467 | 1.107 | Yes (England, Scotland) |
| Japan | 1,844 | 2,600 | 1.410 | Yes |
| China | 1,626 | 2,641 | 1.624 | No |
| Germany | 1,995 | 2,223 | 1.114 | Yes |
| Brazil | 1,910 | 2,341 | 1.226 | Yes |
| South Korea | 1,700 | 2,018 | 1.187 | Yes |
| India | 2,190 | 2,541 | 1.161 | No |
| France | 1,706 | 1,923 | 1.127 | Yes |
| Australia | 1,025 | 1,244 | 1.214 | Yes |
| Italy | 1,124 | 1,314 | 1.169 | No |
| **Total international** | **72,390** | **85,017** | **1.174** | |

https://github.com/jdiestra1977/FIFA_worldCup_2026_risk

# Appendix B: Supplementary Results

## Parameter uncertainty direction

Figure 5 shows the directional asymmetry of Monte Carlo uncertainty for each disease across all venue cities. The axes compare the upside ratio ($\Lambda_{\text{hi}}/\Lambda_{\text{median}}$) against the downside ratio ($\Lambda_{\text{median}}/\Lambda_{\text{lo}}$). Points that fall above the dashed diagonal have CIs that extend further above the median than below it—these diseases may be underestimated at the central parameter values. Points below the diagonal carry CIs that compress below the median, indicating conservative upper bounds.

Dengue and influenza consistently fall above the diagonal because their central detection rates sit near the lower end of the literature range. Pertussis sits well below the diagonal: the central $\rho = 0.10$ is the *upper* bound of its range, so the plausible space is almost entirely downward. Malaria and measles occupy intermediate, roughly symmetric positions.

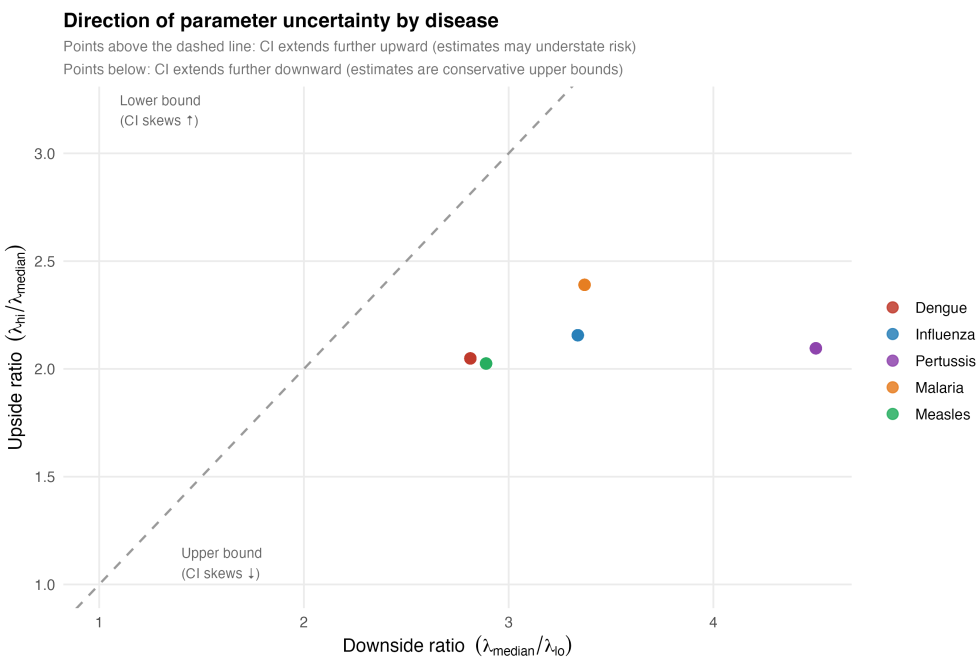



Figure 5: Direction of Monte Carlo parameter uncertainty by disease. Each point represents one disease–city pair under the schedule-driven model. Upside ratio = $\Lambda_{\text{hi}}/\Lambda_{\text{median}}$; downside ratio = $\Lambda_{\text{median}}/\Lambda_{\text{lo}}$. Points above the dashed diagonal: CI skews upward (central estimate near lower bound of plausible range). Points below: CI skews downward (central estimate near upper bound). Dengue and influenza may be underestimated; pertussis estimates are conservative upper bounds.

## Importation risk overview: Baseline and WC-adjusted models

Figures 6 and 7 show the same $P(\geq 1)$ heatmap as Figure 1 in the main text, but for the Baseline (Model 1) and WC-adjusted (Model 2) tiers respectively. City and disease ordering are identical to Figure 1 to allow direct visual comparison across all three model tiers. Dengue importation probability is near-certain ($> 0.99$) at the highest-risk cities under all three models, confirming that this finding is not an artefact of the schedule-driven travel assumptions. The step from Model 1 to Model 2 raises probabilities modestly for most disease–city pairs; the step to Model 3 (main text) introduces the schedule-based concentration at specific match venues.

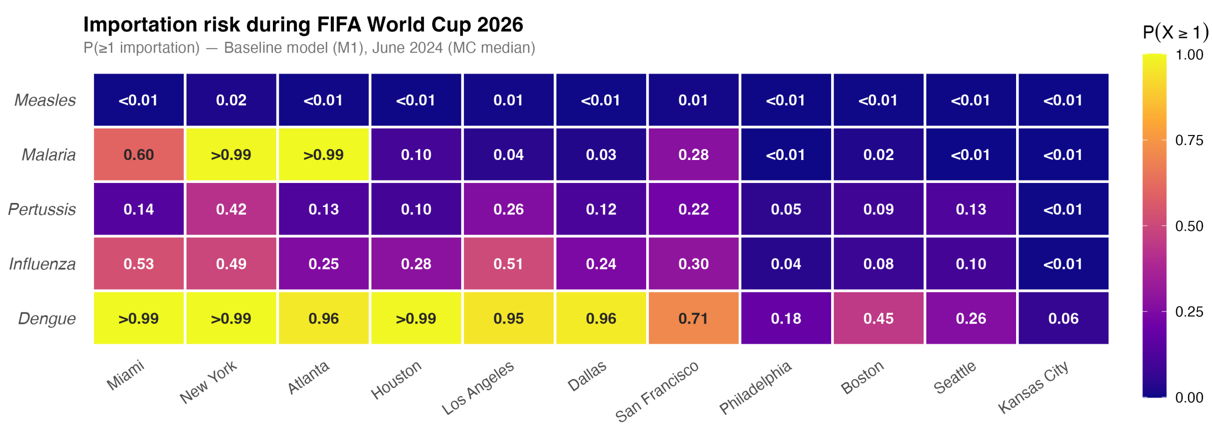


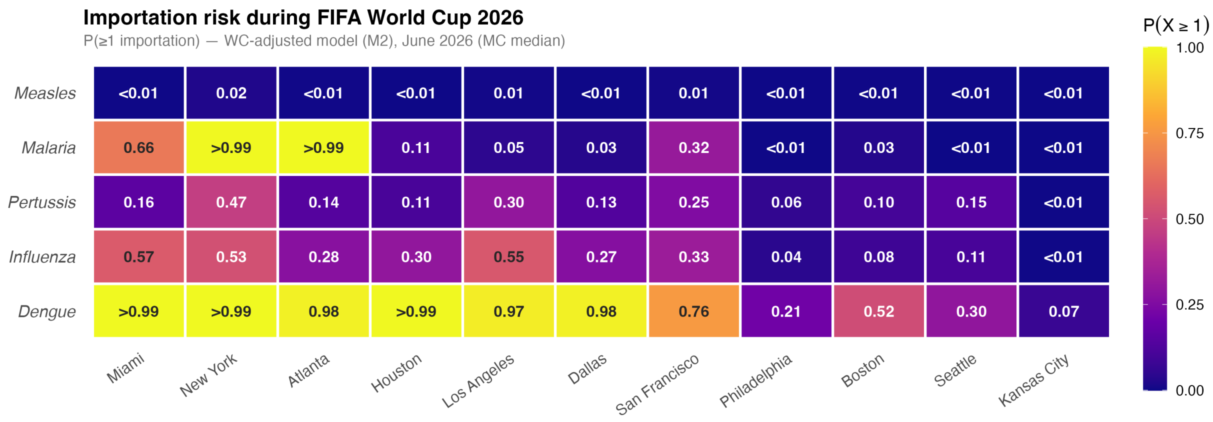

Figure 6: Importation risk overview under the Baseline model (Model 1): $P(\geq 1 \text{ importation})$ using routine June 2024 COR arrivals with no World Cup growth adjustment (MC median, 5,000 draws). Same city and disease ordering as Figure 1.

## Three-model comparison of importation intensity

Figure 8 compares expected importation intensity ($\Lambda$, median and 95% MC uncertainty interval) across the three nested model tiers for the six highest-risk US cities. The near-uniform step-up from Model 1 to Model 2 reflects proportional amplification driven by the WC growth factor; the further step to Model 3 introduces modest city-specific divergence through schedule-based fan routing, without changing city or disease rankings.

## Dengue stream decomposition by source country

Figure 9 shows the dengue-specific travel stream decomposition for the top 20 source countries: background travel (blue) vs. WC fans (orange). The full five-disease stream panel is Figure 10.




| | Miami | New York | Atlanta | Houston | Los Angeles | Dallas | San Francisco | Philadelphia | Boston | Seattle | Kansas City |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Measles | <0.01 | 0.02 | <0.01 | <0.01 | 0.01 | <0.01 | 0.01 | <0.01 | <0.01 | <0.01 | <0.01 |
| Malaria | 0.66 | >0.99 | >0.99 | 0.11 | 0.05 | 0.03 | 0.32 | <0.01 | 0.03 | <0.01 | <0.01 |
| Pertussis | 0.16 | 0.47 | 0.14 | 0.11 | 0.30 | 0.13 | 0.25 | 0.06 | 0.10 | 0.15 | <0.01 |
| Influenza | 0.57 | 0.53 | 0.28 | 0.30 | 0.55 | 0.27 | 0.33 | 0.04 | 0.08 | 0.11 | <0.01 |
| Dengue | >0.99 | >0.99 | 0.98 | >0.99 | 0.97 | 0.98 | 0.76 | 0.21 | 0.52 | 0.30 | 0.07 |

Figure 7: Importation risk overview under the WC-adjusted model (Model 2): $P(\geq 1 \text{ importation})$ using June 2026 projected arrivals scaled by NTTO growth factors (MC median, 5,000 draws). Same city and disease ordering as Figure 1.

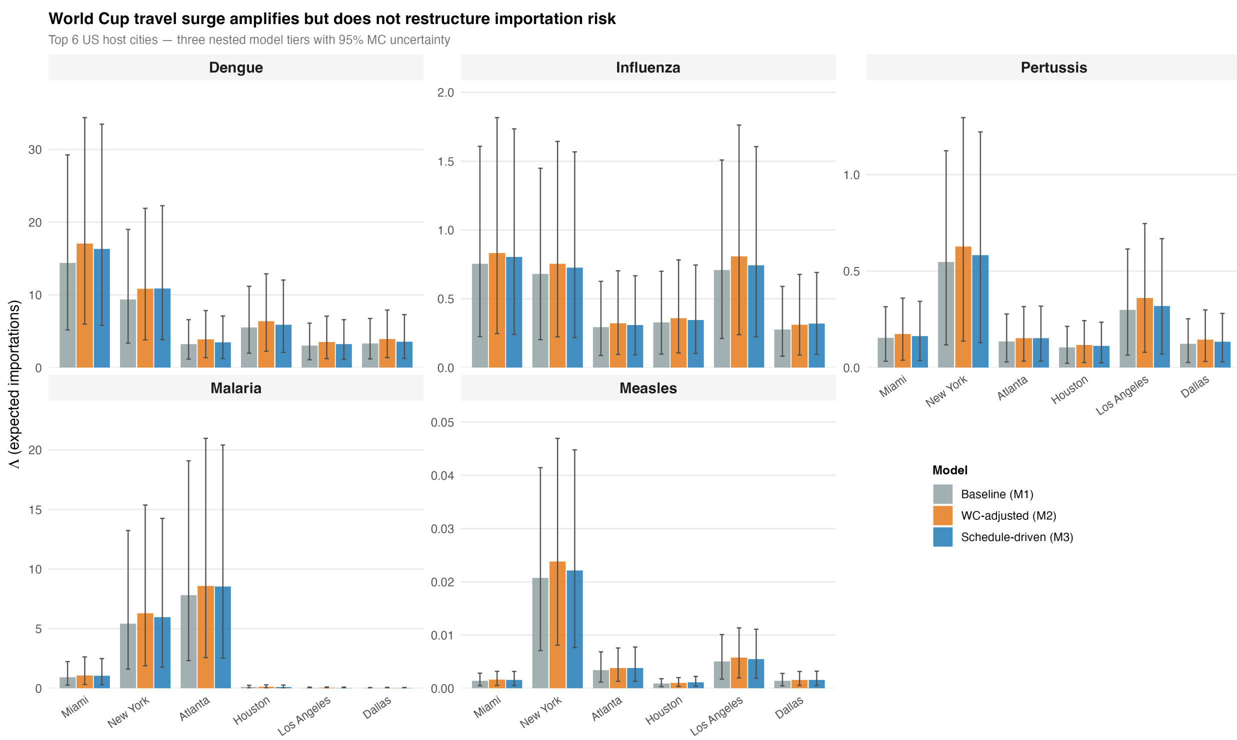



Figure 8: Three-model comparison of expected importations ($\Lambda$) for the six highest-risk US venue cities. Each panel shows one disease; bars are grouped by city and filled by model tier: Baseline (Model 1), WC-adjusted (Model 2), and Schedule-driven (Model 3). Bars show the MC median; whiskers span the 95% uncertainty interval (5,000 Uniform draws on $\rho_d$ and $p_d$). The WC surge amplifies risk proportionally; schedule-based routing introduces modest city-specific divergence but leaves rankings unchanged.

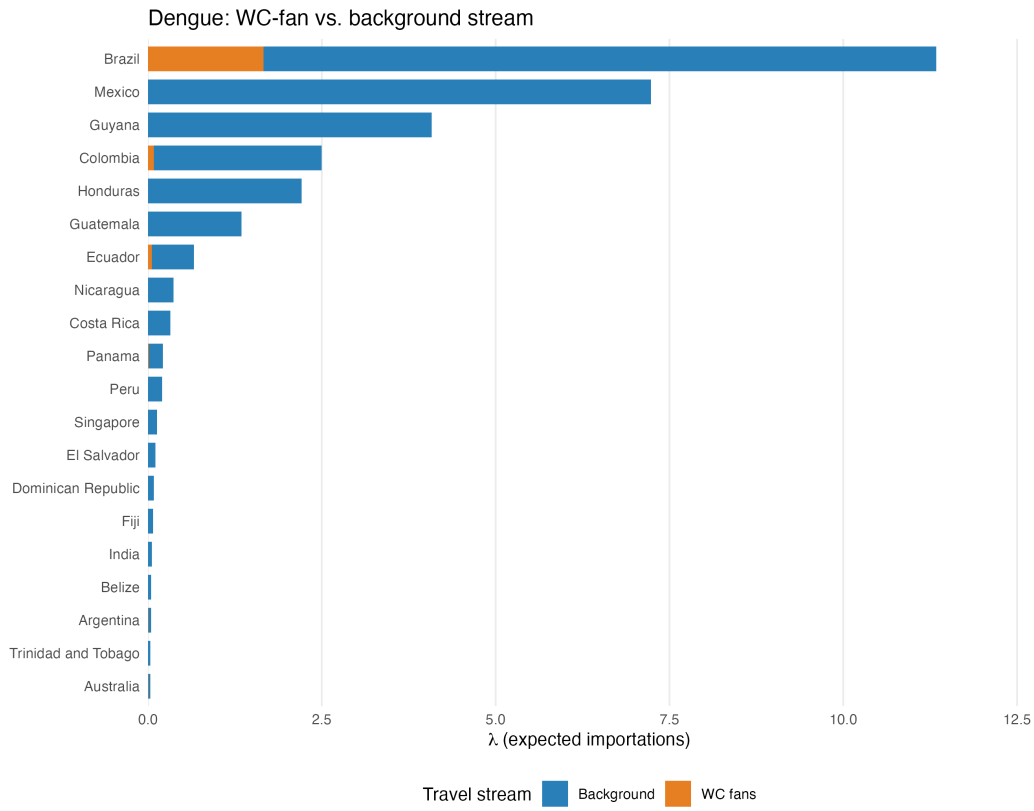



Figure 9: Dengue: importation intensity by travel stream for the top 20 source countries. Blue = background travel; orange = WC fans. Background dominates for all countries; the WC fan increment is largest for high-incidence WC-qualified nations (Brazil, Colombia, Ecuador).

# WC-fan vs. background travel stream decomposition

Figure 10 decomposes the top 20 contributing countries by travel stream (WC fans vs. background) for each of the five diseases. For WC-qualified countries, the WC-fan stream contributes a meaningful but typically minority share of the total importation intensity, consistent with the $\approx 8.3\%$ fan fraction derived from the growth factor decomposition. Countries with the largest WC fan contributions are those that (1) have a qualified team, (2) have high disease incidence, and (3) have their team's matches concentrated at a small number of venue cities, increasing the local fan density at those venues.

Non-qualified countries, regardless of their disease burden, contribute exclusively through the background stream; their presence in the top 20 reflects high June travel volumes to the United States rather than any WC-specific effect. This decomposition highlights that the background stream dominates total importation risk for most source countries: the WC tournament amplifies risk at specific cities and for specific diseases, but baseline travel patterns set the underlying risk floor across all cities and diseases.

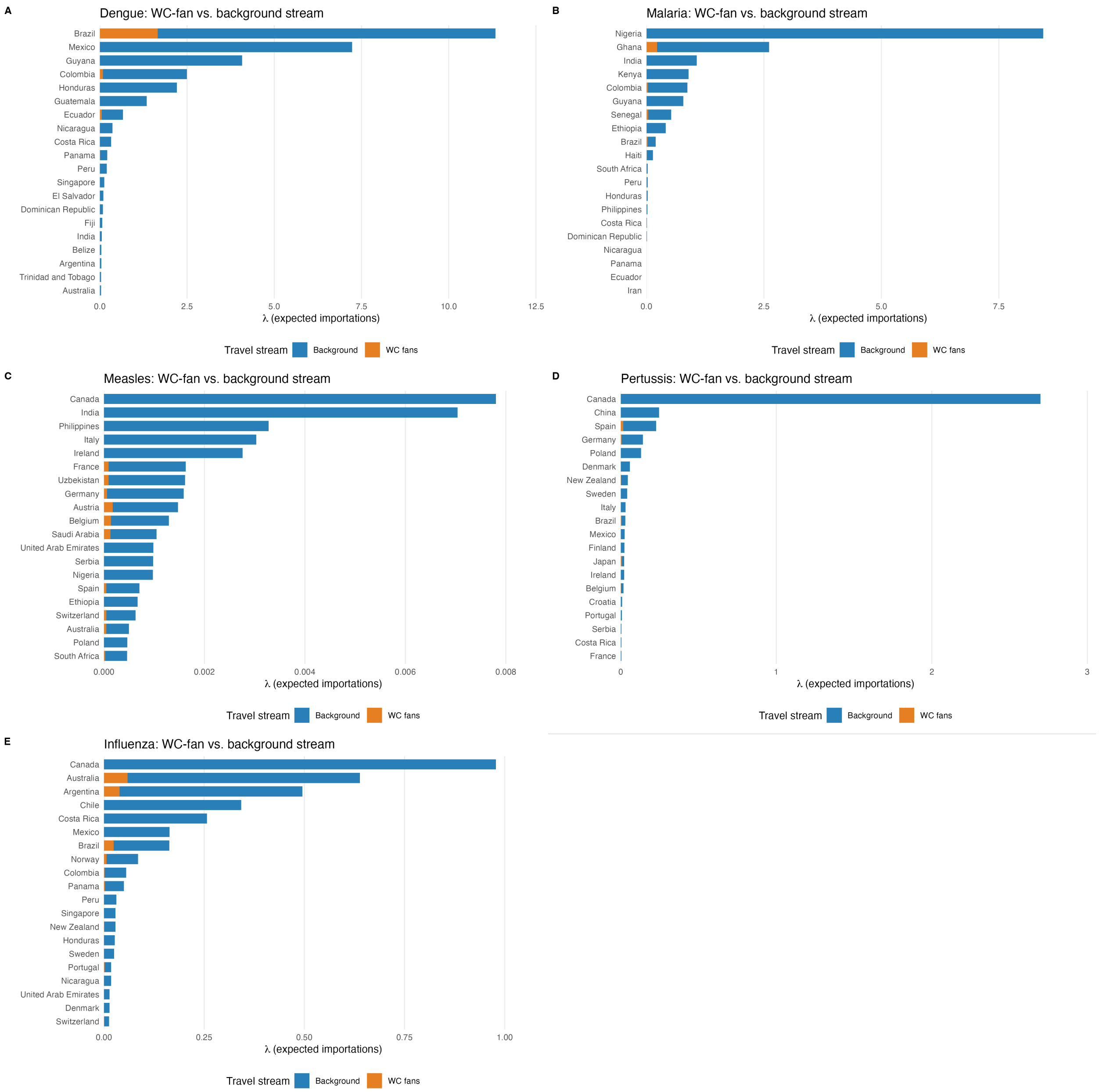



Figure 10: WC-fan vs. background travel stream decomposition for the top 20 source countries by total importation intensity, for each of the five diseases. Orange = WC fans; Blue = background. Countries with WC teams show a non-zero WC-fan component; non-qualified countries contribute only through background travel.

## Country × city importation heatmaps

Figures 11–15 present the country × city importation matrix ($\lambda_{c,v,d}$) for each disease. The heatmaps reveal a consistent spatial pattern: WC-qualified countries from high-incidence regions show importation concentrated at the venue cities where their team plays, producing a block-diagonal structure aligned with the match schedule. Non-qualified countries show importation spread more diffusely across cities in proportion to T-100 routing fractions, reflecting their reliance on normal tourist routing patterns.

For dengue (Figure 11), the highest country–city cells correspond to Latin American WC teams (Brazil, Colombia, Ecuador, Paraguay, Uruguay, Argentina) at their respective match venues. Miami, Houston, and Los Angeles receive notable dengue contributions from multiple Latin American and Caribbean source countries through both the WC-fan and background streams. For malaria (Figure 12), the risk matrix is sparser, with high values concentrated in West African nations at their match venues and lower diffuse contributions elsewhere.

Measles (Figure 13) shows a distinct pattern from dengue and malaria: despite having the highest $\rho$ value, the low $p_d$ suppresses absolute importation intensities across all country–city pairs, resulting in a more uniformly low-intensity matrix with a few notable peaks at cities hosting matches for countries with documented measles outbreaks. Pertussis (Figure 14) shows the broadest geographic spread, consistent with its global endemicity and high $p_d$; the matrix includes non-trivial values from European and East Asian countries in addition to Latin American ones.

# Appendix C: Data Limitations and Improvement Opportunities

## The role of non-public data in improving importation risk estimates

The framework presented here relies entirely on publicly available data sources — a deliberate choice that ensures reproducibility and allows the model to be updated by any public health team with internet access. However, several categories of proprietary and restricted-access data exist that would substantially improve the precision, timeliness, and scope of the estimates, and their absence represents the primary structural limitation of the current analysis.

The most valuable single dataset for this purpose would be **airline booking and ticketing data**. Companies such as OAG (Official Aviation Guide), Amadeus Travel Intelli-

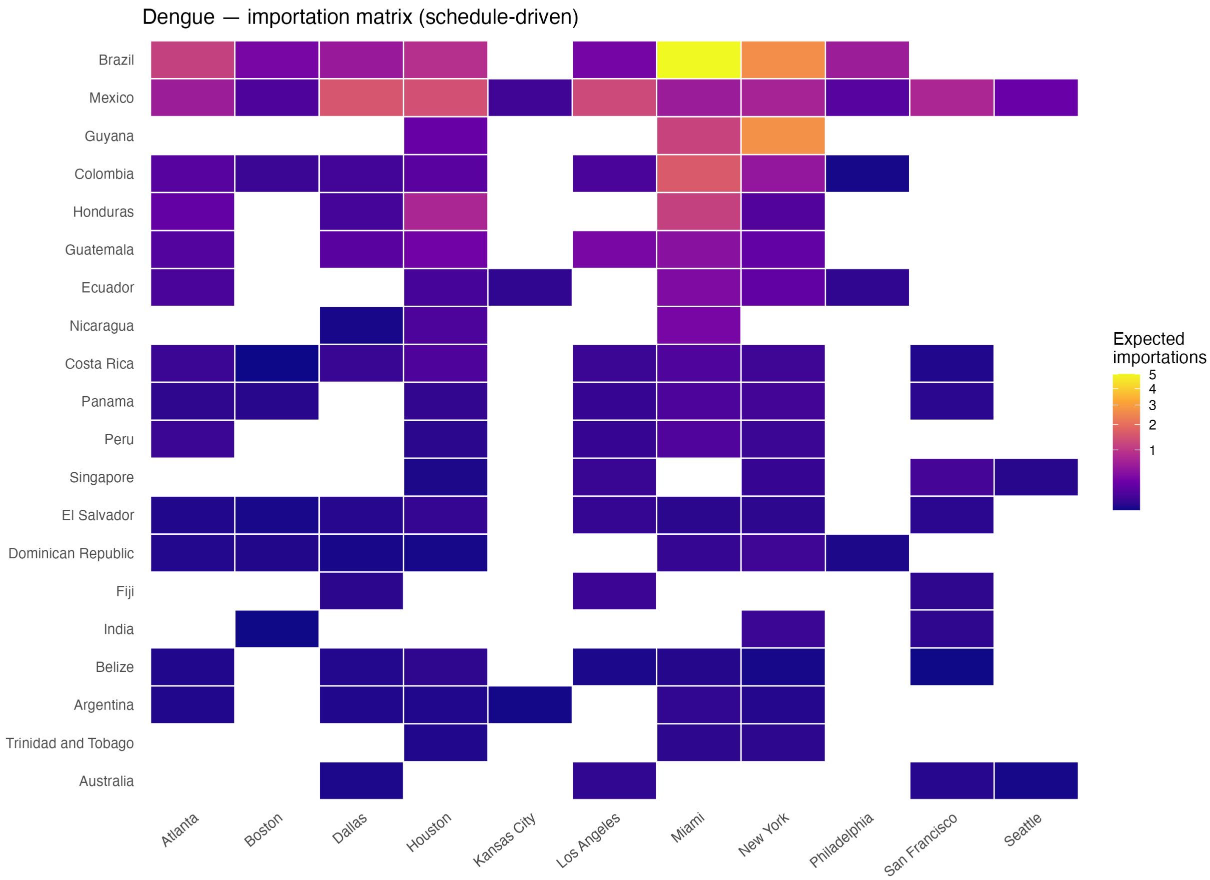



Figure 11: Dengue importation matrix: expected importations $\lambda_{c,v}$ by source country (rows, ordered by total burden) and destination city (columns) under the schedule-driven model. Plasma colour scale: bright yellow/orange = high expected importations; dark purple/blue = low.

gence, and IATA's PaxIS programme maintain origin-to-destination itinerary-level passenger records that cover not only the nonstop international segment (as T-100 does) but the full trip, including domestic connecting legs to non-gateway cities such as Kansas City and Seattle. These records are updated in near-real time and can be filtered to the specific June–July 2026 travel window rather than relying on historical routing fractions as a proxy for future behavior. Similarly, **FIFA ticket sales records by country of purchase**, while not publicly released, would replace the entire WC-fan travel decomposition framework with a direct, event-specific measure: rather than inferring WC fans from NTTO growth factors, one could count the actual tickets sold to fans from each country and allocate them to venue cities by the printed match schedule.

Beyond travel data, the precision of disease burden estimates is constrained by the reliance on **WHO surveillance data**, which is subject to reporting lags, inconsistent national

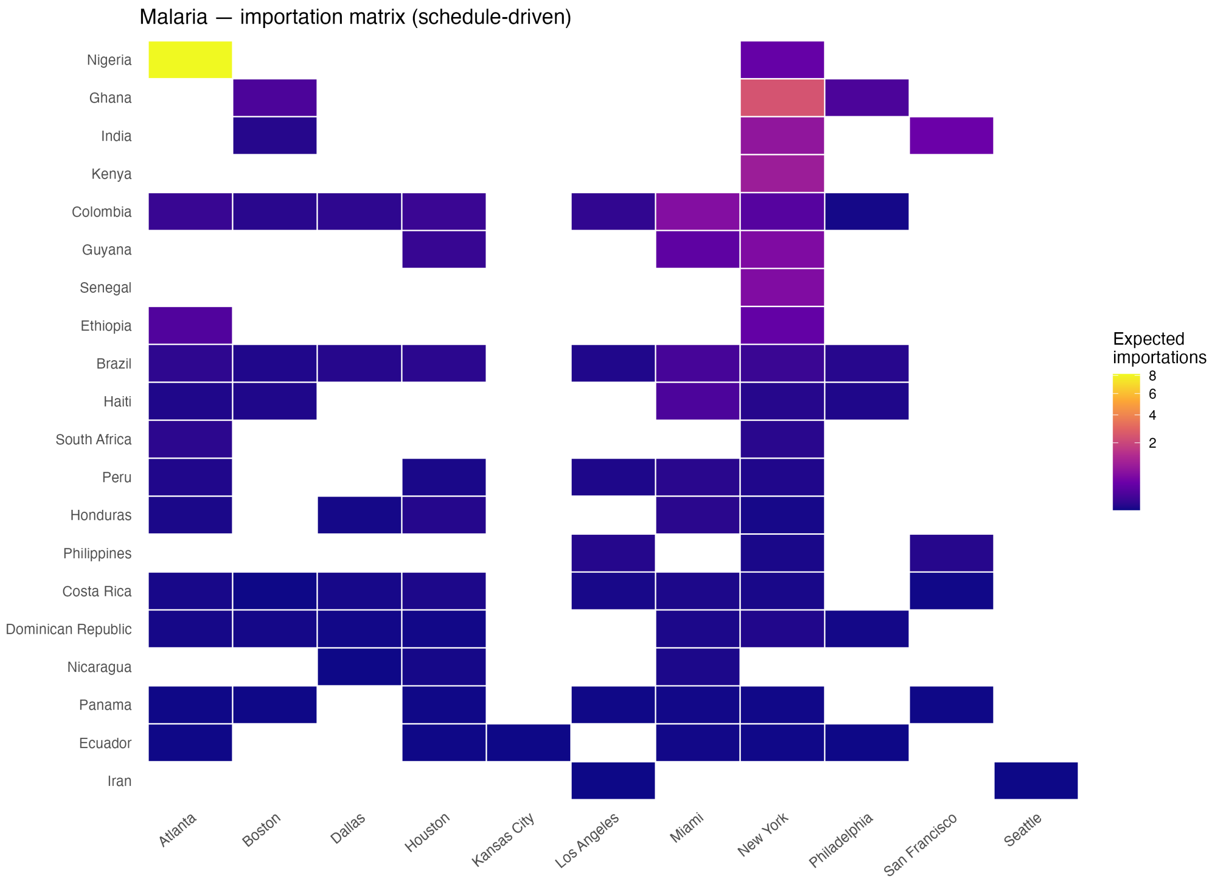



Figure 12: Malaria importation matrix. Same structure as Figure 11. Risk is concentrated in West and Central African source countries.

case definitions, and known under-ascertainment. National or regional health ministry **line-list or case-count data** shared through WHO regional offices, the Pan American Health Organization (PAHO), or the European Centre for Disease Prevention and Control (ECDC) often contain more current and complete figures than what appears in the publicly indexed datasets used here — particularly for dengue, where the 2025 season is not yet fully incorporated. For measles and pertussis, **age-stratified incidence and national vaccination coverage data** at the subnational level (available from national immunization programmes but not systematically published) would substantially improve the per-traveler infection probability $I_{c,d}$ by accounting for the fact that international travelers skew toward working-age adults, who may have markedly different immunity and infection rates than the general population.

Finally, the current model estimates importation risk but stops short of estimating local transmission risk after importation. Extending the framework in that direction would require **local *Aedes aegypti* vector surveillance data** from state and county health

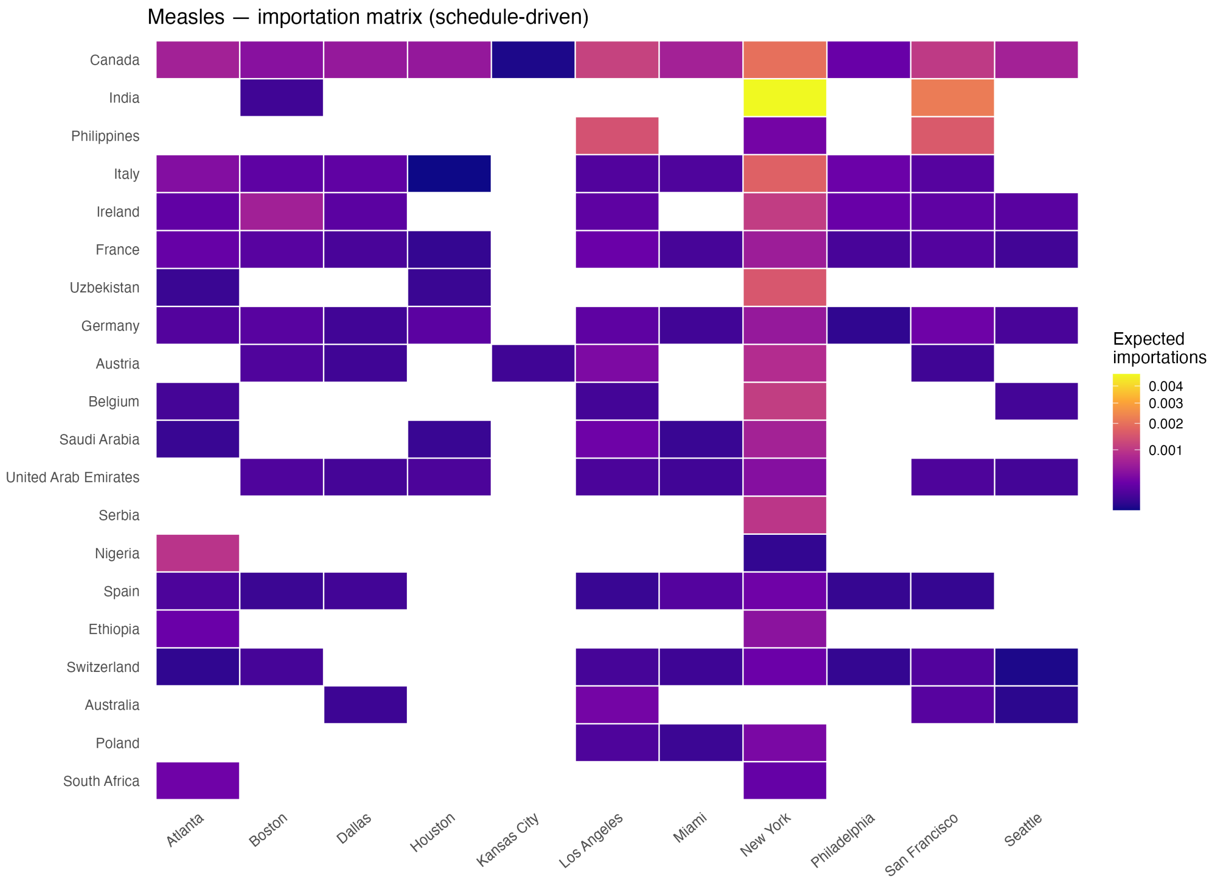



Figure 13: Measles importation matrix. Low absolute intensities across all cells reflect the low travel-while-infectious probability ($p = 0.05$).

departments (partially available through CDC ArboNET but not consistently in machine-readable form), **hospital and public health capacity data** by city and county (available through the American Hospital Association survey but behind a licensing agreement), and **aggregated mobility data** from mobile device providers (Meta, Google) to calibrate the spatial spread of fans within and between cities. None of these datasets are currently accessible through open public APIs at the geographic resolution needed for this analysis. Establishing data-sharing agreements with the relevant agencies and commercial providers before the tournament begins — ideally through the CDC's International Importation Response Branch or the Council of State and Territorial Epidemiologists (CSTE) — would allow these estimates to be refined and operationalized in real time during June–July 2026.

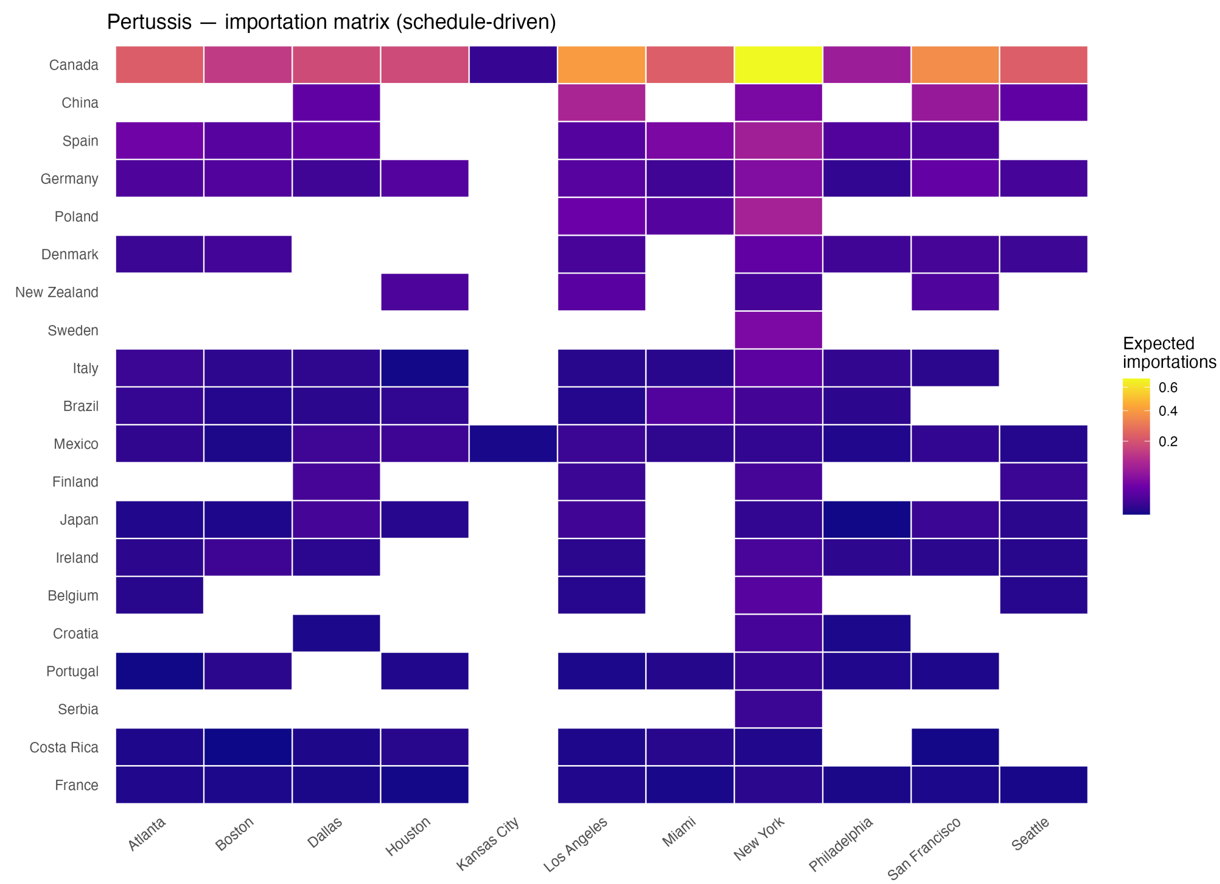



Figure 14: Pertussis importation matrix. The broad geographic spread reflects pertussis's global endemicity and high travel-while-infectious probability ($p = 0.70$).

# Appendix D: Qualified Nations — 2026 FIFA World Cup

| **Country** | **Confederation** | **Host nation** |
|---|---|---|
| Japan | AFC | No |
| Iran | AFC | No |
| South Korea | AFC | No |
| Australia | AFC | No |
| Qatar | AFC | No |
| Saudi Arabia | AFC | No |
| Uzbekistan | AFC | No |

*Continued on next page*

| Country | Confederation | Host nation |
|---|---|---|
| Jordan | AFC | No |
| Iraq | AFC | No |
| Morocco | CAF | No |
| Tunisia | CAF | No |
| Egypt | CAF | No |
| Algeria | CAF | No |
| Ghana | CAF | No |
| Cape Verde | CAF | No |
| Senegal | CAF | No |
| South Africa | CAF | No |
| Ivory Coast | CAF | No |
| DR Congo | CAF | No |
| Canada | CONCACAF | Yes |
| Mexico | CONCACAF | Yes |
| United States | CONCACAF | Yes |
| Panama | CONCACAF | No |
| Curaçao | CONCACAF | No |
| Haiti | CONCACAF | No |
| Argentina | CONMEBOL | No |
| Brazil | CONMEBOL | No |
| Ecuador | CONMEBOL | No |
| Paraguay | CONMEBOL | No |
| Uruguay | CONMEBOL | No |
| Colombia | CONMEBOL | No |
| England | UEFA | No |
| France | UEFA | No |
| Croatia | UEFA | No |
| Portugal | UEFA | No |
| Norway | UEFA | No |
| Germany | UEFA | No |
| Netherlands | UEFA | No |
| Switzerland | UEFA | No |
| Scotland | UEFA | No |

*Continued on next page*

| Country | Confederation | Host nation |
|---|---|---|
| Spain | UEFA | No |
| Austria | UEFA | No |
| Belgium | UEFA | No |
| Bosnia and Herzegovina | UEFA | No |
| Sweden | UEFA | No |
| Turkey | UEFA | No |
| Czech Republic | UEFA | No |
| New Zealand | OFC | No |

*Note*: Confederation abbreviations — AFC: Asian Football Confederation; CAF: Confederation of African Football; CONCACAF: Confederation of North, Central America and Caribbean Association Football; CONMEBOL: South American Football Confederation; UEFA: Union of European Football Associations; OFC: Oceania Football Confederation. Source: `https://en.wikipedia.org/wiki/2026_FIFA_World_Cup_qualification` (accessed April 2026).

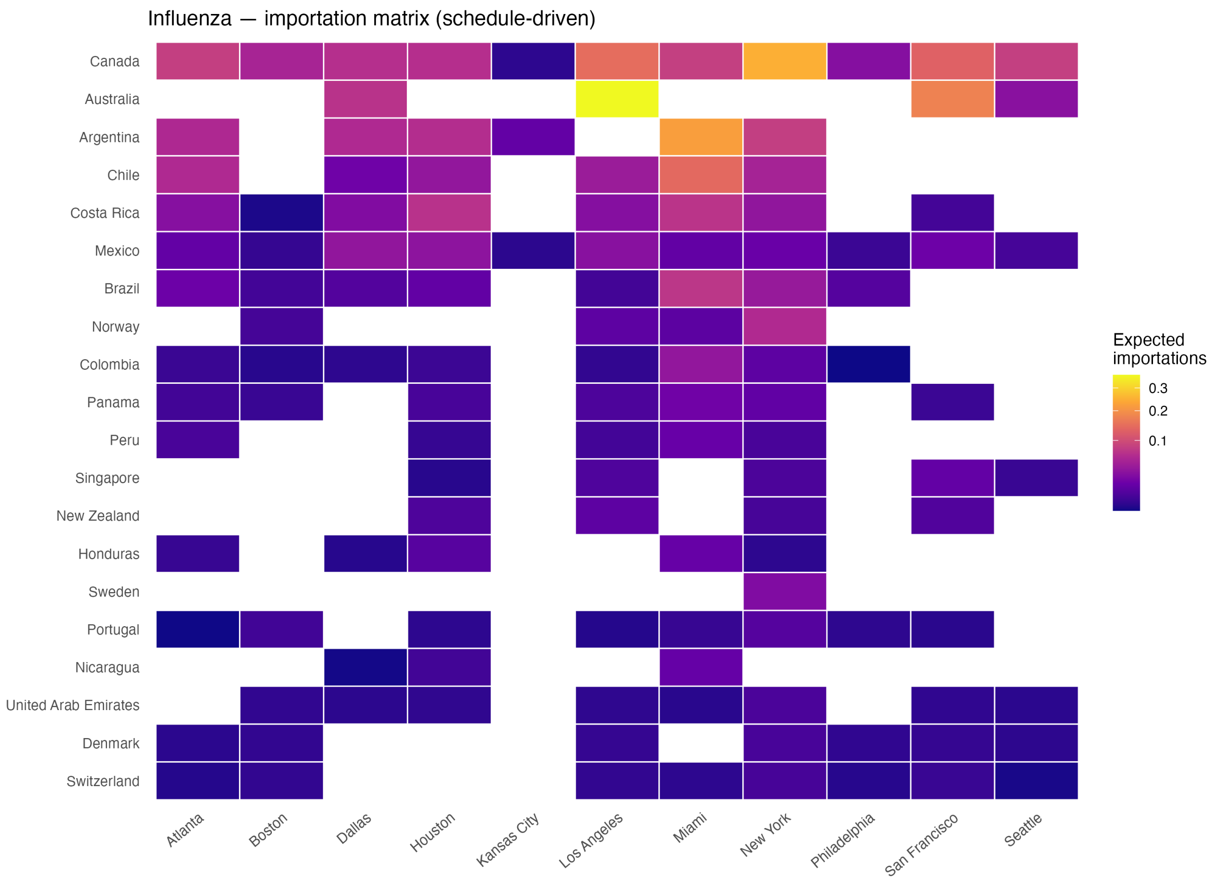



Figure 15: Influenza importation matrix: expected importations $\lambda_{c,v}$ by source country (rows, ordered by total burden) and destination city (columns) under the schedule-driven model. Plasma colour scale: bright yellow/orange = high; dark purple/blue = low. Risk is highest for Southern Hemisphere nations (Brazil, Argentina, Colombia) whose June peak transmission coincides with the tournament window. Background travel from high-volume non-qualified nations (e.g., Canada, Mexico, United Kingdom) contributes a broad diffuse signal across all cities.